Distilling Runtime Safety into VLA Robot Policies

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Abstract

Vision-language-action (VLA) models generalise across manipulation tasks but carry no notion of physical safety: on the SafeLIBERO benchmark, $\pi_{0.5}$ displaces a non-target obstacle in 82.5% of episodes. The standard remedy is a runtime control-barrier-function shield, which projects each proposed action onto a safe set. It works, but it is permanent — consuming inference budget at every control step, requiring obstacle geometry at deployment, and adding a component that can itself fail.

This thesis asks whether that corrective behaviour can instead be amortised into the policy's weights. Distilling $\pi_{0.5}$ on its own shielded rollouts reduced collisions to 19.2% *with no shield at inference*, while raising task success from 58.3% to 82.5% on held-out initial states. One distillation round recovers most of the shielded baseline's collision avoidance (80.8% against 86.7%), and the shield finds materially less to correct when reattached, firing on 18% of control steps against 36%.

The transfer is bounded, and the bound is the contribution. A geometry-privileged motion planner, supplying corrections of equal density under the same shield, produced no measurable improvement under a matched comparison; filtering for successful episodes alone produced none either. What transfers is correction at states the policy itself reaches. Four alternative mechanisms — scalar-reward reinforcement learning, language-specified safety, best-of-N selection and generative uncertainty monitoring — were tested and rejected, each for an identified reason.

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Chapter 1

Introduction

1.1 Opening

Vision-Language-Action (VLA) models introduce a paradigm shift in the field of robotics, serving as primary driver towards general-purpose embodied intelligence. By extending the capabilities of Vision-Language Models (VLMs) into physical execution, VLAs unify multimodal perception and low-level control into single end-to-end architectures, replacing classical modular pipelines.

However, deploying VLA policies into real-world, human-centric environments introduces severe safety risks. While hallucinations in pure language models merely yield incorrect text, ungrounded actions or spatial errors in VLAs directly cause physical collisions, property damage, or catastrophic injury.

The aim of this work is to determine whether the safety gains provided by external inference-time mechanisms can be in turn internalised in the model weights, and thus making the VLA inherently safer.

This chapter starts by providing background context, before presenting the research problem, rationale, scope, and significance of the study.

1.2 Background

Robot control was until recently built from modules: perception fed a state estimator, which fed a planner, which fed a controller, each hand-designed and each requiring the interfaces between them to be specified. Vision-language-action models collapse that pipeline into a single network trained end-to-end, mapping camera images and a natural-language instruction directly to motor commands [2]. Pretraining on web-scale vision-language data and large cross-embodiment robot datasets has made these models strikingly general: the policy studied here, $\pi_{0.5}$, performs long-horizon manipulation in homes it has never seen [11].

That generality is not the same thing as safety, and the reason is structural. A VLA is trained to reproduce demonstrated behaviour, so nothing in its objective penalises contact with an object that is not the target, and nothing in the data reliably demonstrates avoidance — human demonstrators rarely produce the near-collisions that would teach it. The consequence is measurable: on the benchmark used throughout this work, an unmodified $\pi_{0.5}$ disturbs a non-target obstacle in more than eight of every ten episodes. Throughout this thesis, *safety* refers narrowly to this property — avoiding unintended physical contact with objects in the workspace — rather than to the broader alignment concerns associated with language models.

This has not gone unaddressed, and the responses fall into three groups. The largest places a corrective component between the policy and the robot, checking each proposed action against a safety condition — analytic or learned — and modifying it where the condition is violated [8, 10, 14]. A second changes what the policy optimises, expressing safety as a constraint on the learning problem rather than as a filter on its output [15, 17]. A third distils a corrective signal into the policy's weights, so that safer actions are produced rather than intercepted [16]. Purpose-built evaluation has followed: SafeLIBERO augments an established manipulation suite [9] with obstacles placed deliberately in the robot's path [8], and is the setting for everything reported here. The first group is much the most developed, and is where this thesis begins.

That group rests on a control-theoretic answer to constraint satisfaction that predates the learning literature. A *control barrier function* defines a safe region of the state space and derives, from it, a condition on the robot's velocity that keeps the system inside that region: satisfy the condition at every instant, and a system that starts safe provably remains safe [1]. Because that condition is linear in the commanded velocity, it can be enforced by a small optimisation solved at each control step, which takes whatever action a controller proposes and returns the nearest action that satisfies the constraint. Such a component is commonly called a *shield* or *safety filter*: it sits between the policy and the robot, passing safe actions through unchanged and minimally correcting unsafe ones.

Applying this machinery to vision-language-action models is recent. The immediate prior work — the same work that contributes SafeLIBERO — inserts exactly such a filter between $\pi_{0.5}$ and the robot, grounding obstacle geometry from the model's own observations, and demonstrates a large reduction in collisions while preserving the policy's ability to follow instructions [8]. Runtime shielding is therefore the current state of the art for making a VLA safe — and the starting point for the problem this thesis addresses.

1.3 Problem Statement

That a learned policy can be made safer by an external component is established. A barrier-function filter placed between $\pi_{0.5}$ and the robot substantially reduces collisions on SafeLIBERO while preserving instruction-following [8], and learned variants achieve similar ends without an analytic barrier [10, 14]. Separately, safety-relevant behaviour has been shown to transfer into a policy's weights: by imitating a robust barrier-based expert under stated conditions [4], by collecting corrections at the states a learner's errors produce [3, 7], and by distilling a corrective signal into a vision-language-action policy directly [16].

What has not been established is whether the two can be joined. Every filtering method retains its corrective component at deployment, solving an optimisation or running a gradient correction at each control step. Every transfer result either requires conditions no large visuomotor policy satisfies [4], or distils from a teacher constructed out of the student itself [16]. Whether the corrections a control-theoretic shield produces can be compiled into a policy's weights, such that the shield can be *removed*, has not been asked.

The gap matters because the filter is a mitigation rather than a solution, and because its benefit and its cost arise from the same mechanism. It is imperfect: even in the strongest published configuration on this benchmark, roughly a quarter of episodes still end with an obstacle displaced [8]. And it interferes — the authors of that work report that enforcing a constraint can drive a policy into states its training never covered, where it may behave erratically and fail to recover [8]. Correcting actions at runtime is simultaneously what makes the filter effective and what pushes the policy out of distribution, so the two cannot be separated by tuning: a filter that intervenes less interferes less and protects less.

Internalisation is attractive because it does separate them. If the corrections shape the policy's behaviour during training, their effect can survive without a correction occurring at deployment — and the resulting policy can still be shielded where a safety case demands a runtime guarantee, so the two mechanisms are complementary rather than exclusive. The question is therefore not only whether the shield can be removed, saving the inference budget it consumes and the obstacle geometry it requires at deployment. It is whether the safety a runtime filter supplies can be had without the interference it imposes.

1.4 Rationale

Research aim. This research aims to determine whether the collision avoidance supplied by an inference-time control barrier function shield can be amortised into the weights of a vision-language-action policy, such that the shield is no longer required at deployment.

Research objectives. The following objectives guide the study:

1. To implement a CBF shield over $\pi_{0.5}$ and quantify its effect on collision and task success against the published baseline for the SafeLIBERO benchmark.
2. To distil the shielded policy's own corrected rollouts into its action head, and to measure the resulting policy's safety and capability with the shield removed at inference.
3. To isolate the mechanism responsible for any transfer, through controlled ablations that vary the source of the demonstrations and the presence of the shield during collection.
4. To characterise the limits of the transfer: which collisions persist and why, and whether alternative mechanisms achieve the same end.

Research questions. These objectives translate into four questions, each addressed in a corresponding section of Chapter 4:

RQ1. Does a control barrier function shield make a pretrained vision-language-action policy safe on this benchmark, and how do the resulting figures compare with published results? (Section 4.3)

RQ2. Can that safety be internalised by distillation, such that the policy remains safe with the shield removed, and at what cost to task capability? (Section 4.4)

RQ3. What determines whether the transfer occurs — what mechanism produces it, and does the source of the demonstrations matter? (Sections 4.5 and 4.6)

RQ4. What does the transfer fail to cover, and do alternative mechanisms succeed where it does not? (Sections 4.7 and 4.8)

1.5 Scope

This study is conducted entirely in simulation, on the `SafeLIBERO` benchmark. That setting was chosen because it is where the runtime-shielding baseline this work builds on was itself evaluated, which makes the comparison in Chapter 4 direct rather than approximate. No physical hardware was used and no claim of sim-to-real transfer is made.

The policy under study is $\pi_{0.5}$, operating in an operational-space action representation with a flow-matching action head. A single policy was chosen deliberately: the central comparison is between demonstration sources for the same student, and varying the architecture at the same time would confound the two.

The barrier is constructed from obstacle geometry read directly from the simulator rather than estimated from the policy's observations. This is a deliberate isolation rather than a convenience. Estimated geometry would introduce perception error into the corrections being distilled, and a failure to transfer could then be attributed either to the learning problem or to the noise in the supervision. Removing perception error makes the result interpretable, at the cost that the shielded figures reported here are an upper bound for this class of filter.

Within that setting, safety carries the narrow meaning given in Section 1.2. The obstacles are static and rigid, so the study does not address moving obstacles, human proximity, contact-force limits, or self-collision — each of which would need a different barrier and, in some cases, a different sensing arrangement.

1.6 Significance

The practical case for this work rests on how learned manipulation policies are actually deployed. A vision-language-action model cannot be certified by inspection, so any deployment near people or expensive equipment places a safety component in front of it — and that component then becomes part of the system that must be justified, maintained, and accounted for when it fails. This study offers three findings that bear directly on that arrangement. Distilling before shielding produces the lowest collision rate of any configuration measured, at a task success rate indistinguishable from shielding the base policy, so where a runtime constraint is required it is better applied to a distilled policy than to a raw one. The distilled policy also requires the shield roughly half as often, which matters wherever motion must be predictable: a filter that fires constantly is a filter that makes cycle times vary. And safe demonstrations are approximately twice as cheap to collect with the shield active than without it, which is a direct cost saving for any deployment gathering task-specific data.

None of this means a vision-language-action policy is ready to run unshielded on a factory floor. What it establishes is that the mechanism works: safety supplied by an external filter can be moved into a policy's weights, in a setting where that filter is the current state of the art, and the gains survive removal of the filter on held-out conditions. That makes the remaining work characterised rather than open-ended — a barrier covering the whole robot body rather than principally the end-effector, geometry estimated from perception rather than read from a simulator, and validation on physical hardware in the target environment.

Academically, the study addresses a question the literature has left open. Runtime filtering and learned safety representations both retain a corrective component at deployment, and the results showing that safety can be transferred into weights either require conditions large visuomotor policies do not satisfy or distil from teachers derived from the student itself. This work joins the two, and in doing so establishes a boundary condition: the transfer depends on where the supervision sits rather than on who produced it, a distinction the prior literature predicts but does not test. A secondary contribution is measurement rather than method — within a given state, the safety outcomes of a policy's sampled action candidates prove to be almost perfectly correlated, which explains why selection among actions cannot substitute for correcting them.

1.7 Structural Overview

Chapter 2 reviews the literature in three strands: runtime safety filtering for learned policies, whether through an analytic barrier or a learned one; work on what makes corrective demonstrations transferable into a policy's weights; and policy-level alternatives that pursue safety through constrained learning rather than through correction at all. It sets these on the control-theoretic account of safety as forward invariance and the theory of covariate shift in imitation learning. It closes by locating the gap this study addresses and stating precisely what is and is not new about it.

Chapter 3 describes the benchmark and metrics, the geometric construction of the barrier and the shield built from it, the two demonstration sources that are compared, and the distillation procedure. Chapter 4 reports the experiments, organised by research question rather than chronologically, and includes the controls used to isolate the mechanism and the alternative approaches that were tested and rejected.

Chapter 5 interprets those results against the literature of Chapter 2, establishing what each finding contributes and where it sits against prior work. Chapter 6 returns to the research questions and states what the study answers, the limitations that bound those answers, the research they recommend, and what the study contributes. Supporting material that would interrupt the argument — a standalone characterisation of the scripted expert, and the diagnostics behind one rejected approach — is placed in the appendices.

Chapter 2

Related Work

2.1 Introduction

This chapter reviews the research relevant to the aim stated in Section 1.4: determining whether the collision avoidance supplied by an inference-time control barrier function shield can be amortised into the weights of a vision-language-action policy. Its purpose is to establish what is already known about making learned policies physically safe, to identify what remains unresolved, and to position the study against the work it builds on.

The review covers three bodies of work: runtime filtering, in which an external component corrects a policy's actions as they are produced; methods that transfer safety-relevant behaviour into a policy's parameters through imitation; and policy-level approaches that pursue safety through constrained learning rather than filtering. It excludes formal verification of neural networks, which establishes properties of a fixed network rather than shaping its behaviour, and the alignment literature concerned with language-model outputs, which addresses a different sense of safety from the physical one defined in Section 1.2.

The chapter is organised by approach rather than chronologically. Section 2.2 presents the control-theoretic framework the study instantiates. Sections 2.3.1 to 2.3.3 synthesise the empirical work, moving from methods that keep a corrective component at deployment to those that attempt to remove it. Section 2.4 states the resulting gap and the type of gap it is.

2.2 Theoretical Framework

Two established bodies of theory underpin this study. The first defines what it means for a controller to be safe and supplies the mechanism that enforces it. The second determines whether behaviour enforced by that mechanism can be transferred into a policy by supervised learning.

2.2.1 Safety as Forward Invariance

Section 1.2 introduced the barrier informally; this section gives the form the shield of Chapter 3 instantiates. A *control barrier function* encodes a safe set $\mathcal{C}$ as the superlevel set of a differentiable function h, and converts safety into a condition affine in the control input: any u satisfying $\dot{h}(x,u) \geq -\alpha(h(x))$ renders $\mathcal{C}$ *forward invariant*, meaning a system that begins inside the set never leaves it [1].

Because that condition is affine in u, it can be imposed as one linear inequality per constraint inside a quadratic program that stays as close as possible to a nominal input:

$$u^{\star} = \arg\min_{u} \tfrac{1}{2}\|u - u_{\mathrm{nom}}\|^2 \quad \text{s.t.} \quad \nabla h^{\top}(f + gu) \geq -\alpha(h). \tag{2.1}$$

This is the object the thesis calls a *shield*: a filter that accepts whatever a nominal controller proposes and returns the minimum modification satisfying the constraint.

Two properties make it usable as a teacher rather than merely as a guard. It is agnostic to how the nominal action was produced, so it composes with a policy it knows nothing about. And because it minimises $\|u - u_{\mathrm{nom}}\|$, its output is not an arbitrary safe action but the *closest* safe action to the one the policy wanted — which is precisely the form a supervised learning target should take.

That second property is why this formulation was adopted over the alternatives. A constrained Markov decision process expresses safety as a bound on expected cost and yields an optimised policy, not a per-step correction, so it produces nothing to imitate. A reward penalty expresses safety as a scalar added to the objective, leaving the policy to infer which of several hundred actions was responsible for a cost it incurred. Only the barrier formulation returns a corrected *action*, at the step where the correction was needed, in the space the policy emits.

2.2.2 Imitation and Covariate Shift

Whether such corrections transfer is a separate question, governed by a different theory. *Behaviour cloning* treats imitation as supervised learning: minimise the discrepancy between the learner's action and the demonstrator's, over a dataset of demonstrated state-action pairs. Its weakness is well characterised. The learner is trained under the demonstrator's state distribution but evaluated under its own, and because control is closed-loop, small errors move it to states the demonstrator never visited, where its error is unconstrained and compounds — a failure mode that grows quadratically rather than linearly in the task horizon [12].

The established remedy is to change where the supervision is collected rather than how much of it there is: query the expert at the states the *learner* visits, and aggregate those labels into the training set [12]. Applied to safety specifically, the same idea motivates querying a safe controller precisely when the learner is about to need it [18].

This is the theoretical basis for a prediction the study tests directly. If what makes corrective supervision transferable is the state distribution it occupies, then two demonstration sets of equal size and equal correction density should differ in effect according to where their states sit. Section 4.6 constructs exactly that comparison.

2.2.3 Two Limits on What May Be Claimed

The guarantee of Equation 2.1 is continuous-time and assumes known control-affine dynamics, so a discrete-time implementation on a contact-rich manipulator satisfies it only approximately (Section 3.5). More fundamentally, forward invariance is a property of the *closed loop containing* the quadratic program, not of the policy inside it. It is therefore not a quantity that can be extracted and stored in weights: remove the program and the theorem's premise fails, whatever the policy has learned. This is why the question of Chapter 4 is empirical rather than a corollary, and why its result is reported as a measured reduction in collisions rather than as a guarantee.

2.3 Empirical Research

2.3.1 Filtering at Runtime

The most direct route to a safe learned policy is to place a corrective component in front of it, and the strongest evidence that this works comes from the study that introduced the benchmark used here. AEGIS inserts a safety-constraint layer between $\pi_{0.5}$ and the robot: a vision-language module grounds obstacle geometry from observations, and a CBF-QP modifies the policy's proposed action whenever it violates the resulting constraint [8]. The effect is large — collision avoidance rises from under 19% to between 69% and 78% depending on the action space, while task success improves rather than degrades — and the cost is modest, at 0.356 ms per control step.

Learned variants pursue the same end without an analytic barrier, and their differences are instructive. ConBaT trains a safety critic from binary safe/unsafe labels and operates in embedding space rather than state space, so it applies where explicit geometry is unavailable; at deployment it rectifies actions by backpropagation through that critic until the barrier conditions hold [10]. Latent Policy Barrier instead treats the latent manifold of expert demonstrations as an implicit barrier, training a dynamics model on both expert data and the policy's own rollouts, then steering in latent space at inference [14]. The two differ in what defines the safe set — a learned critic against a demonstration manifold — but converge on the same deployment architecture.

That convergence is the pattern worth extracting. Across all three, safety is supplied by a component that runs at inference and leaves the policy itself unmodified. Latent Policy Barrier makes this explicit, advertising plug-and-play compatibility with pretrained policies without retraining or fine-tuning. The consequence is that the question this thesis asks cannot be posed within this line of work: removing the component returns the base policy by construction, because nothing was ever transferred into it.

2.3.2 Transferring Safety into Weights

A second body of work does modify the policy, and asks under what conditions safety-relevant behaviour survives in its parameters. Cosner et al. establish that it can, formally:

imitating a CBF-based expert yields a learned controller with input-to-state-safety guarantees, provided the expert is a *robust* barrier program with explicit disturbance margins, the dynamics are known and control-affine, and the learned controller satisfies conditions on data density along the safe-set boundary, bounded learning error, and a known Lipschitz constant [4]. Even then the guarantee holds on an expanded safe set that degrades with data sparsity. The demonstrations are low-dimensional — an inverted pendulum and a vehicle on a track — and no large visuomotor policy satisfies the stated conditions. What survives for this thesis is not the theorem but its framing: the goal is to transfer safety *guarantees* rather than to reproduce the expert's behaviour, a distinction Section 4.7 observes empirically.

Where formal conditions are unavailable, the empirical literature converges on a different answer: what matters is not the fidelity of the imitation but the location of the supervision. SAFE-GIL injects reachability-computed adversarial disturbances during collection, steering a fixed expert toward the states a learner's errors would produce so that its recovery behaviour is recorded [3]. Its ablation is the decisive evidence: Gaussian noise, uniform noise and DART all fail to reproduce the benefit, so it is the *targeting* of the disturbance rather than augmentation as such that carries the effect. IntervenGen arrives at the same conclusion by a different route, rolling out the current policy so that recorded mistakes are its genuine failures, then transforming human recovery segments onto the states reached, and reporting that ten interventions expanded synthetically outperform a hundred collected directly [7].

These two agree on a point that is easy to misstate. In SAFE-GIL the expert is an MPC or PID controller; in IntervenGen the recovery actions come from a human teleoperator. Neither teacher is the learner, and in both cases safety transfers. What the two share is not the identity of the teacher but where its demonstrations sit.

That reading matters because the nearest work in the vision-language-action setting appears at first to contradict it. ROAD-VLA distils a proximal teacher — constructed by perturbing the student's own action-token logits with calibrated advantage estimates — along the student's on-policy rollouts, and derives a policy-improvement bound that holds only while the teacher remains close to the current policy [16]. Its privileged-teacher variants fail badly, which is often read as evidence that external teachers cannot transfer. The variants it rejects are, however, all *textual*: a retrieved action hint, an egocentric spatial description, and a task plan. The teacher it accepts is derived from the student itself. No foreign low-level controller appears anywhere in that work, so it neither supports nor refutes the state-coverage account — it simply does not test it.

One result does bear directly on the protocol this thesis adopts, and it is a negative. Guerrier et al. train a reinforcement-learning agent with a CBF filter active throughout training and then remove it [6]. Safety is not retained: the agent collides once the filter is disabled, and its critic assigns high value *inside* the obstacle, indicating the hazard was never represented at all. Their agent's actions were overridden by the filter, but its learning signal remained a scalar reward, so no gradient ever pointed toward the corrected action. Their curriculum variant, which blends filter and policy actions with the filter's weight decayed to zero, does internalise the constraint. Taken together this establishes that exposure to a shield during training is not by itself sufficient — what the policy is trained *on* decides the outcome.

2.3.3 Constrained Learning as an Alternative

A third line pursues safety through the training objective rather than through filtering or imitation. SafeVLA formulates the problem as a constrained Markov decision process, eliciting unsafe behaviour and optimising the policy under explicit constraints [17]. Safe-Dojo performs model-based safe reinforcement learning inside an action-conditioned video world model, estimating imagined task progress and safety cost separately and optimising them with Lagrangian constrained GRPO; it evaluates on SafeLIBERO, making it the most direct methodological alternative in this literature [15].

These matter as calibration rather than contrast. Neither is scalar-reward reinforcement learning, and both supply substantially richer constraint information than a penalty term. Any negative result this thesis reports for scalar rewards is therefore a result about the signal tested, not about constrained learning as a class.

2.3.4 Methodological Patterns and Open Tensions

Two features of how this literature is evaluated bear on how its findings can be combined. The first is that the reported quantities are not mutually comparable. AEGIS defines collision avoidance as the proportion of strictly collision-free episodes and states no numeric tolerance [8]; SAFE-GIL reports collision rates on two-dimensional navigation and aircraft taxiing [3]; ROAD-VLA reports task success under distribution shift [16]. Sample sizes range from tens of trials to sixteen hundred episodes, and single-seed or few-seed reporting is common. Cross-paper numerical comparison is therefore mostly unavailable, which is why this study reproduces its baseline directly rather than quoting a published figure (Section 4.3).

The second is scale. The transfer results that are best characterised are also the furthest from the setting studied here: multilayer perceptrons on low-dimensional state in SAFE-GIL [3], a unicycle model in a 1.5 m arena in Guerrier [6], an inverted pendulum in Cosner [4]. Whether their conclusions hold for a three-billion-parameter flow-matching policy is an open empirical question rather than a settled one.

Against that, the field agrees on three things. Runtime filtering substantially reduces collisions for learned manipulation policies. Corrective demonstrations transfer more effectively than additional clean ones. And scalar rewards are a poor channel for safety, a conclusion reached independently by studies of CBF-guided reinforcement learning [6] and by the constrained formulations that avoid them [15, 17].

One tension remains unresolved, and it is the one this study addresses. SAFE-GIL and IntervenGen show foreign teachers transferring safety; ROAD-VLA is widely read as showing they cannot. The readings are reconcilable only if the operative variable is where the supervision sits rather than who produced it — and no existing study varies the teacher's state distribution while holding the correction signal fixed. A second, smaller tension concerns combination: SAFE-GIL reports filtering and corrective imitation as complementary without much degradation in task performance, a claim Section 4.4 is positioned to test directly.

2.4 Research Gap

The gap this study addresses is primarily **methodological**, with a secondary **theoretical** dimension. It is methodological because the existing work relies on a single architectural arrangement — safety supplied by a component that runs alongside the policy — and the alternative arrangement, in which the corrections that component produces are compiled into the policy itself, has not been evaluated. It is theoretical because the framework that would predict when such a transfer succeeds has been developed and tested only at a scale far removed from contemporary vision-language-action models.

The methodological gap is structural rather than an oversight, and this distinction matters. In the filtering literature the policy is never modified: AEGIS corrects actions as they are produced [8], ConBaT rectifies them by backpropagation at test time [10], and Latent Policy Barrier steers in latent space while explicitly requiring no retraining [14]. Removing the component from any of these returns the base policy unchanged, so there is no transfer to measure. The question cannot be posed within that line of work.

Where a policy *is* shaped under a filter, the one available result is negative. Guerrier et al. train under an active barrier and then remove it, and find the constraint was never internalised [6]. That study is the closest existing analogue to the present one, and it differs in a single respect: its learning signal was a scalar reward, so the corrected action was imposed on the policy but never presented to it as a target. Whether the same protocol succeeds when the correction is the imitation target has not been tested. The gap is therefore not that nobody looked, but that the one attempt used a channel the corrections could not travel through.

The theoretical dimension follows from Section 2.2. Conditions under which a learned controller inherits safety from a barrier-based expert are established [4], but demonstrated on an inverted pendulum and a vehicle on a track, under assumptions — known control-affine dynamics, bounded learning error, a known Lipschitz constant — that a three-billion-parameter flow-matching policy does not satisfy. Separately, covariate-shift theory predicts that supervision transfers according to the states it occupies [12], a prediction the empirical literature supports indirectly [3, 7] but has never tested by varying the state distribution while holding the correction signal fixed.

This study addresses that gap by constructing the missing arrangement and the missing comparison: distilling a control-theoretic shield's own corrections into the policy that generated them, evaluating the result with the shield removed, and contrasting it against a teacher matched in correction density but not in state distribution. The claim of novelty is correspondingly narrow. Learning safety from barrier-based supervision is not new [4], and distilling a corrective signal into a vision-language-action policy is not new [16]. What has not been done is the conjunction — a control-theoretic filter, distilled from a policy's own shielded rollouts, evaluated with the filter removed, in the setting where runtime shielding is the current state of the art — together with the boundary condition that establishes what the transfer depends on. This study does not resolve the question of how to make vision-language-action policies safe; it establishes that one mechanism transfers, and identifies what governs whether it does.

2.5 Conclusion

This chapter has reviewed three bodies of work bearing on the safety of learned manipulation policies. Runtime filtering, whether through an analytic barrier or a learned one, reliably reduces collisions but supplies safety through a component that runs alongside the policy and leaves it unchanged. Work on transferring safety into a policy's parameters establishes that corrective demonstrations can carry it, and converges on the location of the supervision rather than the identity of the teacher as the operative variable — though the results that characterise this best are drawn from settings far smaller than a contemporary vision-language-action model. Constrained learning offers a third route, pursuing safety through the training objective rather than through correction at all. Underpinning the first of these is the control-theoretic account of safety as forward invariance; underpinning the second is the theory of covariate shift in imitation learning.

The gap that follows is methodological, with a theoretical dimension. No study has taken the corrections a control-theoretic filter produces, compiled them into the policy that generated them, and evaluated what remains once the filter is removed — and the one study that trained a policy under an active filter used a learning signal through which those corrections could not travel.

Addressing that gap requires three things of the method: a shield whose corrections are substantial enough to constitute a training signal, two demonstration sources matched in correction density but differing in the states they occupy, and a distillation procedure that can be evaluated with the filter both present and absent. Chapter 3 sets out each in turn.

Chapter 3

Method

3.1 Introduction

The aim stated in Section 1.4 is to determine whether the collision avoidance supplied by an inference-time control barrier function shield can be amortised into the weights of a vision-language-action policy. Section 2.5 identified three things a method must supply to test that: a shield whose corrections are substantial enough to constitute a training signal, two demonstration sources matched in correction density but differing in the states they occupy, and a distillation procedure that can be evaluated with the filter both present and absent. This chapter sets out each, and the reasoning behind the design decisions they involve.

The chapter proceeds from the setting to the specific. Section 3.2 states the research design and the variables it holds fixed. Section 3.3 describes the benchmark, the metrics, and the division of initial states into training and evaluation. Sections 3.4 and 3.5 construct the barrier and the quadratic program built from it, and state the approximations each involves. Section 3.6 describes the two demonstration sources whose comparison is the central experiment, and Section 3.7 the procedure that trains on them. Section 3.8 sets out how the resulting measurements are compared.

3.2 Research Design

The study uses a **controlled experimental design**: a variable is manipulated, everything else is held fixed, and the difference in outcome is attributed to the manipulation. The research questions are causal — whether the shield's corrections *cause* a transfer of safety, and what property of those corrections determines it — and only a design that isolates variables can answer them. A descriptive or correlational treatment could establish that shielded rollouts and safer policies coincide, but not that one produces the other.

Four comparisons make up the study, each manipulating a single variable. The first varies whether the shield runs at inference, establishing that its corrections are large enough to constitute a training signal. The second varies whether the policy has been distilled, and measures the result with the shield removed. The third varies the source of the demonstrations while holding their number, filtering criteria and correction density fixed. The fourth varies whether the shield was active during collection, holding the demonstration count and the retention criteria constant.

Across all four, the following are held fixed: the base checkpoint, the training objective and optimiser configuration, the number of gradient steps (matched to within one percent, Section 3.7), the evaluation code, the scenes, and the held-out initial states on which every condition is evaluated. Action sampling is deterministic except where a comparison requires otherwise, so a given initial state produces the same rollout on repetition.

That degree of control is what the simulated setting buys, and it is worth stating as a design choice rather than only as a limitation. Two policies can be evaluated from bit-identical starting conditions, a demonstration set can be subsampled to match another exactly, and a rollout can be repeated. None of that is available on hardware, where initial conditions vary between trials and the same policy produces different trajectories from nominally identical starts. The comparisons in Chapter 4 depend on that precision: the matched control of Section 4.5 would be uninterpretable if the two conditions differed in demonstration count or training length as well as in the presence of the shield.

The design establishes what varies with what, within this benchmark and this policy. It does not establish that the relationships found here hold at other scales, on other architectures, or under perception error, and Section 6.3 states those boundaries.

3.3 The SafeLIBERO Benchmark

All experiments use SafeLIBERO [8], which augments the LIBERO manipulation suite [9] with obstacles placed in the workspace. The grid is three task suites (spatial, object, goal) $\times$ two obstacle levels $\times$ four tasks, giving 24 distinct scenes, each providing 50 randomised initial states.

Experiments run in MuJoCo through the robosuite interface that LIBERO builds on [9]. The platform is a simulated seven-degree-of-freedom Franka Emika Panda with a parallel-jaw gripper, controlled in operational space: the policy emits a seven-dimensional action — a six-degree-of-freedom end-effector pose delta plus a binary gripper command — with each axis bounded to $[-1, 1]$, and an operational-space controller resolves it to joint torques. Observations are a 224×224 agent-view image, a wrist camera at the same resolution, and eight proprioceptive dimensions: end-effector position, orientation as an axis-angle, and the two gripper finger positions. Notably the observation contains no joint angles, a fact Section 4.5 returns to.

Each action chunk is produced by integrating $\pi_{0.5}$'s learned velocity field. Under the linear flow $x_t = t\epsilon + (1-t)a$, the latent at $t = 1$ is a Gaussian sample ϵ and the latent at $t = 0$ is the action a, so generating an action means integrating from one end of that path to the other; ten Euler steps are used, referred to as denoising steps by the convention of the diffusion literature from which flow matching inherits its vocabulary. Two distinct sources of randomness are available. The first is the initial latent ϵ, which seeds generation and is always present. The second is an optional per-step perturbation that converts the sampling ODE into an equivalent SDE; its magnitude is set by a parameter this work calls the *noise level*, and it is zero everywhere except in the reinforcement-learning and best-of-N experiments of Section 4.8. At zero the integrator reduces exactly to the deterministic ODE, so a given initial latent always yields the same action chunk and rollouts are reproducible from a fixed initial state.

Initial states 0–34 are used for demonstration collection and training; states 35–49 are held out and used only for evaluation. The split serves the central claim: a policy trained on shielded rollouts could in principle reproduce those trajectories rather than acquire the behaviour behind them, and evaluating on initial states it has never seen distinguishes the two. The evaluation varies object placement within trained scenes rather than introducing new tasks or objects, so it tests generalisation of avoidance across configurations, not across tasks.

Every reported condition uses five held-out initial states from each of the 24 scenes, giving $n = 120$ rollouts. Sampling a fixed number per scene rather than drawing 120 at random keeps every scene equally weighted, so a suite the policy happens to find easy cannot dominate the pooled figure. The number is bounded by cost: each rollout is a 300-step simulation driven by a three-billion-parameter policy, and the six-condition evaluation of Section 4.4 alone requires 720 rollouts.

3.3.1 Metrics

Task success rate (TSR) is the fraction of rollouts in which the environment's own goal predicate is satisfied at any point before the horizon expires. Success is binary per rollout.

Collision is recorded when the obstacle's position departs from its initial position by more than one millimetre in the ℓ_1 sense,

$$\sum_{k \in \{x,y,z\}} \left| p_k(t) - p_k(0) \right| > 0.001 \text{ m}, \tag{3.1}$$

and a rollout is marked collided if this holds at any control step. Collision-avoidance rate (CAR) is the complement.

The reference pose $p(0)$ is taken after a brief settling period, since objects spawned in unsupported poses would otherwise cross the threshold under gravity before the policy has acted.

Per-body attribution records which of the robot's bodies contacted the obstacle in each episode, grouped as gripper, arm link, held object, and other scene objects. The collision metric registers *whether* the obstacle moved, not *what* moved it, and the barrier's authority is uneven across the robot — precise at the end-effector, approximate at the links — so this decomposition is what makes the residual interpretable.

Execution time to success (ETS) is the control step at which the success predicate first holds, averaged over successful rollouts only and reported in control steps. ETS is therefore conditional on success and must be read alongside TSR.

CBF activation rate is the fraction of control steps on which the shield altered the action, counted whenever the projected action differs from the nominal by more than 10^{-4} in norm. It measures how *often* the shield intervened, not how large the corrections were, and serves as an independent check on internalisation: a policy that has learned to avoid obstacles should need fewer corrections when the shield is reattached.

Mean correction norm is the average $\|u^\star - u_{\text{nom}}\|$ over steps where the shield intervened, measured against actions bounded to $[-1, 1]$ per axis. Where the activation rate says how often the shield acted, this says how far it moved the action when it did. The two are independent: a shield could intervene constantly with negligible corrections, or rarely with large ones.

3.4 Obstacle Geometry and the Barrier

The obstacle's collision mesh is read from the simulator, sampled into a point cloud, and decomposed into a small set of spheres. Sphere decomposition is used rather than a single bounding volume because a bounding sphere over a non-convex object is a poor approximation.

End-effector. Three spheres in the end-effector frame approximate the Franka hand: a palm sphere of radius 4.8 cm at 5.6 cm behind the grasp site, and two finger-pad spheres of radius 2.0 cm offset 3.6 cm either side at 10.5 cm depth.

Arm links. Links three through seven and the palm body are each assigned a radius (5.5 cm to 9.0 cm, set from the link's maximum transverse extent plus a 15–21 mm margin) and sampled at three positions along the link axis, so a long link is not represented by its origin alone.

For each pair of a robot sphere i and an obstacle sphere j the barrier is

$$h_{ij} = \|p_i - c_j\|^2 - (r_i + r_j)^2, \tag{3.2}$$

with $h > 0$ the safe set. Differentiating along the rigid-body velocity gives the constraint row

$$\big[\, 2\, s\, (p_i - c_j)^\top R_1 \,\big]\, u \;+\; k\, h_{ij} \;\geq\; 0, \qquad k = 10, \tag{3.3}$$

where u is the commanded end-effector velocity in the body frame, R_1 its rotation and s a scale factor. For arm links, whose velocity is not the commanded one, the mapping $v_{\text{link}} \approx J_{\text{link}} J_{\text{ee}}^{+}\,(sR_1 u)$ is used, with J_{ee}^{+} a damped pseudo-inverse ($\lambda = 10^{-3}$).

Five properties of this construction bear on Chapter 4.

- *The robot model is approximate* The end-effector is covered by three fitted spheres; each arm link by three point samples carrying one radius. The end-effector is represented more faithfully than the links.
- *Links one and two are not constrained*, being close to the base and rarely near the obstacle in these scenes.
- *Arm-link velocities are estimated, not commanded.* The QP optimises an end-effector velocity and infers each link's motion through a damped Jacobian pseudo-inverse. Where that inverse is ill-conditioned, the predicted link velocity and the realised one diverge, and the constraint is enforced against the prediction.
- *The geometry is privileged.* Obstacle poses and sizes are read from the simulator rather than estimated from the policy's observations, a scope decision justified in Section 1.5. The figures reported here are therefore an upper bound for this class of filter.
- *Arm links are constrained here but not in the originating work*, whose formulation solely constrains the end-effector and which notes that its unconstrained kinematic links may consequently collide [8]. Extending the barrier to the links is a deliberate difference from that baseline, and Section 4.3 reports its effect.

3.5 The CBF Shield

At each control step the policy's proposed translation is treated as a nominal end-effector velocity u_{nom}, and the shield solves

$$u^\star = \arg\min_u \|u - u_{\text{nom}}\|^2 \quad \text{s.t.} \quad a_m^\top u + b_m \geq 0 \;\text{ for every constraint row } m, \tag{3.4}$$

a quadratic program in three variables whose rows are the end-effector/obstacle and arm-link pairs of Section 3.4. This is the standard CBF-QP form [1], instantiated on the sphere-pair barrier of Equation 3.2. When no constraint is active the solution is u_{nom} and the action passes through unchanged.

The problem is solved with OSQP [13] through CVXPY [5]. An intervention is recorded when the returned action differs from the nominal by more than 10^{-4}.

Two discrete-time adjustments are required because the barrier's guarantee is continuous-time. The effective radius is inflated by a lag buffer covering the distance travelled between control steps and the inverse-kinematics tracking error. Rotation is not constrained: the barrier acts on the translational channel only.

3.6 Teachers

Two sources of demonstrations are compared, and the comparison between them is the central experiment of this work.

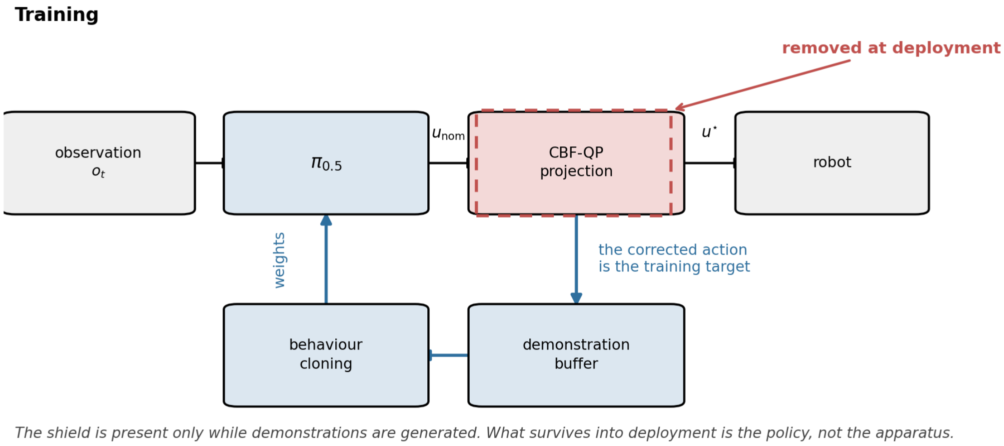

Figure 3.1: The shield during demonstration collection. The policy proposes an action, the quadratic program projects it onto the safe set, and the robot executes the result — while the corrected action is tapped off as the imitation target. At deployment the projection block is removed and the policy runs alone.

3.6.1 The Shielded Policy as Its Own Teacher

The policy is rolled out with the shield active, and episodes are retained if they succeeded, displaced nothing, and the shield intervened at least once. That third criterion matters: a clean episode in which the shield never fired is ordinary base-policy behaviour and teaches nothing about safety.

3.6.2 A Privileged Scripted Expert

The alternative teacher is a sampling-based motion planner: RRT-Connect in joint space with clearance inflation, followed by inverse kinematics and playback as operational-space deltas. It reads object and goal poses directly from the BDDL scene specification [9].

The planner produces collision-free plans, but not collision-free executions: inverse-kinematics tracking error and operational-space control cause the realised trajectory to depart from the planned one. Its demonstrations are therefore collected under the same shield as the policy's own (Section 3.5). The two teachers differ in the states their demonstrations occupy, not in whether they were corrected — the planner plans from its own initial conditions and is never conditioned on where $\pi_{0.5}$ fails.

3.6.3 Why $\pi_{0.5}$, and What Was Tried First

The work began on OpenVLA-7B with a Cartesian CBF filter, and the first distillation attempt collected CBF-corrected demonstrations and fine-tuned that model through LoRA. It produced no useful learning signal.

OpenVLA-7B's base competence on LIBERO is low enough that safety and capability failures are difficult to separate: a policy that rarely completes the task provides little opportunity to observe whether it completes it *safely*.

$\pi_{0.5}$ has substantially stronger base LIBERO performance, which makes the safety question well-posed, and it is the policy the runtime-shielding baseline evaluates [8], which makes the comparison in Section 4.3 direct.

3.7 Distillation

Retained demonstrations are behaviour-cloned using the model's native flow-matching loss [2, 11]. Only the parameters selected by the training configuration's trainable filter receive gradients — a LoRA adapter on the action head — while the vision-language backbone is frozen. The imitation target is the sequence of actions actually executed, that is post-shield, so the policy learns the corrected behaviour rather than its own original proposals. Actions are normalised by passing them through the policy's own input transform, so the target matches the distribution the model was pretrained on rather than a hand-rolled normalisation. Because the corrections are applied to actions the policy itself proposed, they are located by construction at states the policy reaches — the property Section 2.2 identifies as governing whether supervision transfers.

The procedure sits in the DAgger lineage [12], and shares with SafeDAgger [18] the idea that a safe controller's interventions are themselves the informative signal.

Training Control and Alignment When comparing distilled policies, training runs are matched strictly on the total number of gradient steps rather than the number of epochs. This controls for disparities in episode lengths across demonstration sources: a shielded VLA rollout over a horizon of 300 steps yields approximately 35 training samples, whereas a scripted motion planner episode over a 900-step horizon yields approximately 412 samples. Matching by epoch count would result in severe step imbalances (16,140 vs. $\approx$ 212,000 gradient steps). The distilled policies compared are matched at 16,140 and 15,988 gradient steps respectively (a variance of $< 1\%$).

3.8 Analysis

Unless a caption states otherwise, every headline quantity is a proportion over $n = 120$ independent held-out rollouts, reported with a 95% Wilson score interval. Wilson is used rather than the normal approximation because several conditions sit near zero or one — shielded gripper collisions are zero across 360 episodes — and at these rates the normal interval is too narrow and can extend outside $[0, 1]$.

Comparisons are made by interval overlap: a difference is reported as established when the two intervals are disjoint, and as suggestive when they overlap. This is deliberately conservative. Disjoint intervals imply a significant difference, but overlapping intervals do not imply the absence of one, so the criterion understates rather than overstates the evidence. Where it matters — the arm-link comparison of Section 4.7 — the overlap is stated explicitly rather than left for the reader to notice.

Results are pooled across the 24 scenes rather than reported per scene. Pooling is appropriate because the manipulated variable is the same in every scene and the interest is in its average effect; per-scene reporting at five rollouts each would produce intervals too wide to distinguish anything. Where the per-scene distribution carries information that the pooled figure hides, it is shown separately, as in the coverage comparison of Section 4.6.

Two secondary analyses support the main comparisons. Per-body collision attribution (Section 3.3.1) decomposes the residual by which part of the robot made contact. And demonstration sets are characterised by activation rate and correction norm, so that a claim about what was learned can be checked against what was actually in the training data.

3.9 Conclusion

The design is a controlled experimental comparison in which a single variable is manipulated at a time and the apparatus is otherwise held fixed. That apparatus comprises a benchmark of 24 scenes with a training and evaluation split over initial states, a sphere-decomposition barrier over the end-effector and arm links, a quadratic program that projects the policy's proposed action onto the resulting safe set, two demonstration sources differing in the states they occupy but not in whether they were corrected, and a behaviour-cloning procedure matched across conditions on gradient steps. Measurements are proportions over held-out rollouts, compared by interval overlap.

Each element answers to one of the three requirements set out at the start of the chapter. The shield supplies corrections dense and large enough to constitute a training signal; the two teachers differ in state distribution while matching in correction density; and the distillation target is the corrected action itself, so the policy can be evaluated with the filter both present and absent. Chapter 4 reports what that yields.

Chapter 4

Experiments and Results

4.1 Introduction

A vision-language-action policy that is competent at manipulation is not thereby safe, and the standard remedy — a control barrier function shield correcting its actions at every step — remains attached for the life of the deployment. This study asks whether those corrections can instead be compiled into the policy, so that the shield is no longer required (Section 1.4).

The method is the controlled comparison set out in Chapter 3: a shield is built over $\pi_{0.5}$, its corrections are collected as demonstrations, those demonstrations are behaviour-cloned back into the policy's action head, and the result is evaluated with the shield removed. Every condition reported here is $n = 120$ rollouts on held-out initial states, with proportions given as Wilson 95% intervals and differences read from interval overlap (Section 3.8).

This chapter presents what those experiments produced. It reports the measurements and states what each contributes to the research questions of Section 1.4; the interpretation of what they mean, and how they sit against the literature, is deferred to Chapter 5.

The chapter follows the research questions in order. Section 4.3 establishes whether the shield works at all (RQ1) and Section 4.4 whether its effect survives distillation (RQ2). Sections 4.5 and 4.6 address what determines whether the transfer occurs, first by isolating the shield from outcome selection and then by varying the demonstration source (RQ3). Sections 4.7 and 4.8 report what the transfer does not cover and whether alternative mechanisms succeed where it does not (RQ4).

4.2 The Demonstration Sets

Three demonstration sets were collected, and every distillation result reported in this chapter draws on one of them. Their composition is given in Table 4.1.

Two features of the table bear on results reported later. The shielded set is pervasively corrected: the barrier is active on roughly a third of control steps in the median episode, moving the action by 0.33 against a bound of ± 1. And collecting safe demonstrations

Table 4.1: The three demonstration sets. Yield is the fraction of collection rollouts passing the retention filter. Activation rate and correction norm characterise how much the shield altered the trajectories that were kept.

Set	Demos	Rollouts	Yield	Scenes	Activation	Corr. norm
Shielded self-rollouts[a]	186	480	39%	24	0.32	0.33
Privileged planner[b]	318	864	37%	18	—	—
Unshielded control[c]	85	480	18%	24	0	—

[a] Retained if the episode succeeded, displaced nothing, and the shield intervened at least once. Activation is the median fraction of control steps shielded; the median episode carries 51 interventions, the minimum nine, and none fewer than five. Correction norm is measured against actions bounded to ± 1 per axis.
[b] Retained if the episode succeeded and displaced nothing. Six of the 24 scenes produced no usable demonstrations (Section 4.6).
[c] Collected with the shield switched off and filtered on the same success and displacement criteria; the shield-intervened criterion is inapplicable. Used in Section 4.5, where the shielded set is subsampled to 85 to match.

without the shield is roughly twice as expensive, at an 18% yield against 39%.

4.3 Does the Shield Make $\pi_{0.5}$ Safe?

The first question is whether a control barrier function shield, applied as a runtime filter, makes a pretrained VLA safe on this benchmark at all. Table 4.2 reports the comparison: the shield removes 69 percentage points of collision, with disjoint confidence intervals.

Table 4.2: Safety and task performance for base $\pi_{0.5}$ with and without the CBF runtime shield. Pooled over 24 scenes, $n = 120$ held-out rollouts per condition, Wilson 95% intervals.

Policy	TSR (95% CI)	Collision rate (95% CI)	CAR
Base, no shield	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]	17.5%
Base + shield	71.7% [63.0, 79.0]	13.3% [8.4, 20.6]	86.7%

Success also rises, from 58.3% to 71.7%. That direction is worth noting, because in Section 4.4 the same shield applied to an already-avoidant policy lowers success instead.

Answering RQ1. A control barrier function shield does make a pretrained vision-language-action policy safe on this benchmark: collisions fall by 69 percentage points with disjoint intervals, and task success rises rather than degrades. The corrections it applies are therefore large enough to constitute a training signal, which is the precondition for everything that follows.

4.4 Can the Shield's Safety Be Internalised?

The shielded demonstration set of Table 4.1 was behaviour-cloned into the policy's action head, and the procedure was repeated for a second round.

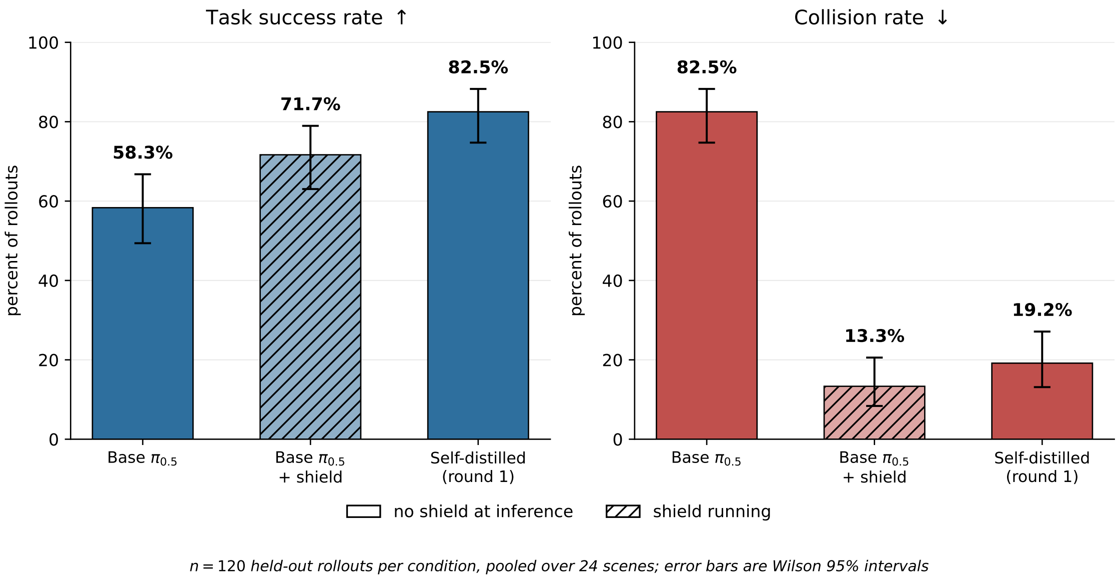

Figure 4.1: The central result. The self-distilled policy, running with *no shield at inference* (solid), is both safer and more successful than the base policy it was distilled from, and recovers most of the safety of the shielded baseline (hatched). Error bars are Wilson 95% intervals at $n = 120$ held-out rollouts per condition.

Table 4.3: Effect of self-distillation, with and without the shield at inference. Rows in bold are the shield-free distilled policies, which are the claim of this work. $n = 120$ held-out rollouts per row, pooled over 24 scenes; intervals are Wilson 95%.

Policy	TSR (95% CI)	Collision (95% CI)	CAR	CBF act.[a]	ETS[b]
Base, no shield	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]	17.5%	—	138.6
Base + shield	71.7% [63.0, 79.0]	13.3% [8.4, 20.6]	86.7%	0.355	160.6
Round 1, no shield	**82.5% [74.7, 88.3]**	**19.2% [13.1, 27.1]**	**80.8%**	—	157.3
Round 1 + shield	70.8% [62.2, 78.2]	8.3% [4.6, 14.7]	91.7%	0.177	170.0
Round 2, no shield	**80.8% [72.9, 86.9]**	**17.5% [11.7, 25.3]**	**82.5%**	—	158.8
Round 2 + shield	72.5% [63.9, 79.7]	9.2% [5.2, 15.7]	90.8%	0.182	176.7

[a] CBF activation rate (Section 3.3.1): the fraction of control steps on which the shield altered the action. Inapplicable when no shield is running, shown as —.
[b] Execution time to success, in control steps, over successful rollouts only (Section 3.3.1). Conditional on success and must be read alongside TSR.

The headline result is the third row. With no shield running at inference, the distilled policy displaces the obstacle in 19.2% of episodes, against 82.5% for the policy it was distilled from — and its success rate rises from 58.3% to 82.5%. Both intervals are disjoint from the base policy's. The distilled policy is simultaneously safer and more capable than its own teacher, without the teacher's filter.

It is worth noting how close this comes to the shielded baseline: CAR 80.8% shield-free against 86.7% with the shield active. Most of the shield's safety benefit survives its removal.

One round is enough. Rounds one and two overlap on both metrics, so the second round buys nothing measurable. It also does not erode performance.

Stacking the shield costs success. Applying the shield to the distilled policy lowers collisions further, to 8.3%, but reduces success from 82.5% to 70.8%. This is the reverse of the effect in Section 4.3, where the same shield raised success. The two observations are reconciled in Section 5.4.

The cost of internalised caution. ETS rises from 138.6 to 157.3 steps, roughly 14% slower. The distilled policy takes a more conservative route.

Answering RQ2. The shield's safety is internalised. With no shield at inference the distilled policy collides in 19.2% of episodes against the base policy's 82.5%, and succeeds in 82.5% against 58.3%, both intervals disjoint. It recovers most of the shielded baseline's collision avoidance (80.8% against 86.7%) without the filter, in a single round, at a cost of roughly 14% in execution time.

4.5 What Explains the Improvement?

Section 4.4 showed that distilling the shielded policy's own clean rollouts makes it dramatically safer without the shield; however, it does not explain why this is so. The demonstrations were filtered on three properties at once — the shield fired, the episode succeeded, and nothing was displaced — and two mechanisms are conflated in the filter.

Under the first, **distillation**, the policy imitates the shield's avoidance corrections. This is the claim of this work. Under the second, **selection**, the improvement comes from training a policy on its own successful episodes, a known self-improvement effect requiring no shield. If the second mechanism accounts for the result, the shield is incidental and the contribution is a rediscovery of filtered behaviour cloning.

To disambiguate this, a matched control was run.

The matched control

The matched control held every variable constant except for switching off the shield when collecting the demonstrations.

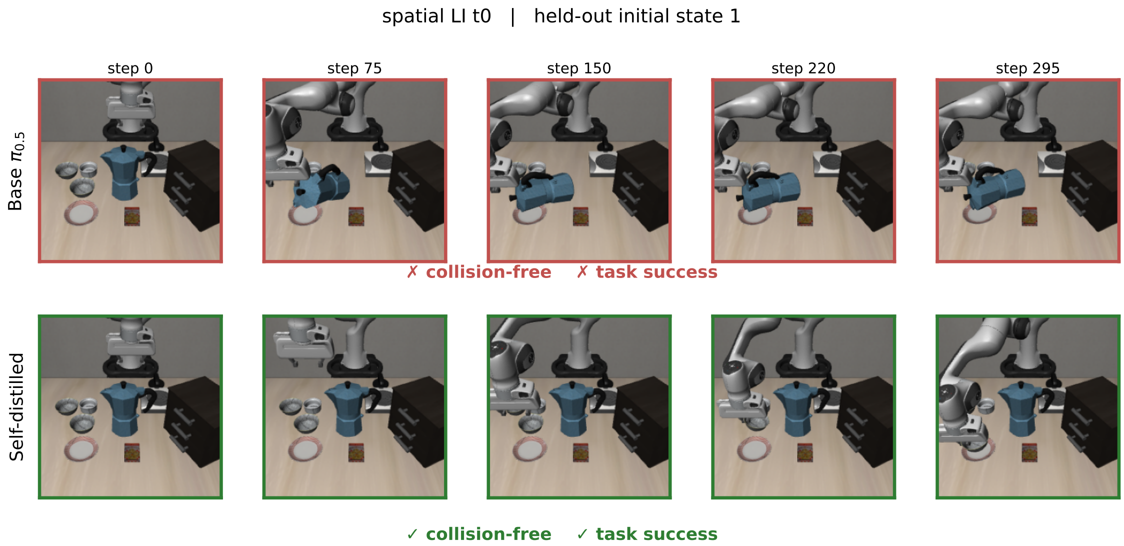

Figure 4.2: Spatial suite, task 0, held-out initial state 1, under the base policy (top) and the self-distilled policy (bottom), neither running a shield. The base policy knocks the moka pot onto its side within the first quarter of the episode and then runs to the horizon without completing the task; the distilled policy leaves the pot standing throughout and succeeds at step 104. Frames are sampled at equal fractions of each episode, so the two rows show the whole of their respective rollouts.

Table 4.4: The matched control. Both distilled conditions use 85 demonstrations passing an identical filter; they differ only in whether the shield was active during collection. $n = 120$ held-out rollouts per row, Wilson 95% intervals.

Condition	Demos	TSR (95% CI)	Collision (95% CI)
Undistilled base	—	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]
Control, shield **off**	85	50.0% [41.2, 58.8]	84.2% [76.6, 89.6]
Matched, shield **on**	85	**75.8% [67.4, 82.6]**	**26.7% [19.6, 35.2]**

Compared to the base policy, the control has a marginally worse collision rate and more substantial TSR degradation. So, success-filtering by itself is unsuccessful in this benchmark. At the same demonstration count and under the same filter, the shielded condition reaches 26.7%, with a confidence interval disjoint from the control's. Therefore, the difference between the two conditions is attributable to the shield. The activation rate corroborates this independently: after distillation the shield finds materially less to correct (Table 4.3), and a policy that merely succeeded more often would not require *less* safety intervention.

The control is not uniformly inert, however. Per-body attribution shows its arm-link collisions falling from 17 to 6 while gripper collisions remain at 44, which is why the total did not move. Outcome selection therefore transfers avoidance in the channel the barrier does not express, while leaving the channel that dominates the collision count unchanged.

Answering RQ3, first part. The improvement is produced by the shield's corrections, not by training on successful episodes. At matched demonstration count and under an identical filter, collection with the shield switched off leaves the policy indistinguishable from the undistilled baseline, while collection with it active produces a disjoint improvement.

4.6 Does the Source of the Demonstrations Matter?

Section 4.5 established that the shield's corrections, rather than outcome selection, produce the transfer. This section asks whether it matters whose rollouts those corrections are applied to. A sampling-based motion planner was used as an alternative teacher: RRT-Connect in joint space with clearance inflation, followed by inverse kinematics and playback as operational-space deltas (Section 3.6.2).

Table 4.5: Comparison of demonstration sources for policy distillation.

Policy	TSR (95% CI)	Collision rate (95% CI)
Base, no shield	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]
Planner-distilled, no shield	54.2% [45.3, 62.8]	80.0% [72.0, 86.2]
Self-distilled (round 1), no shield	**82.5% [74.7, 88.3]**	**19.2% [13.1, 27.1]**

The planner-distilled policy is statistically indistinguishable from the undistilled base policy on both metrics and the only variable is the demonstration source (Section 3.7).

An interactive DAgger variant was also attempted and is reported in Appendix A; it did not alter the conclusion of the matched comparison below.

The planner is not a weaker teacher. It might be supposed that the planner's demonstrations simply contain less safety information. They do not. Run unshielded on spatial task 0 the expert collides in 5 of 8 episodes, with contact from arm links, gripper, held object and scene objects alike; run shielded it collides in none. Its demonstrations depend on the shield to the same degree the policy's own do, and both sets were collected under it.

The demonstration sets are therefore matched in correction signal, and differ in the states they occupy.

The mechanism is state coverage. Figure 4.3 takes the base policy's own visited states and asks how closely each demonstration set approaches them.

Subsampled to equal size per scene, the shielded rollouts cover 37.7% of the policy's own states within 2 cm against the planner's 22.8%, and cover better in 15 of the 18 scenes holding both demonstration types. Without that control the two sets are near-indistinguishable at 37.7% against 32.8%, so the matching is load-bearing rather than cosmetic.

Figure 4.3(b) shows the shape of the difference. The planner is a scripted controller and traces near-identical paths from a given initial condition, so its demonstrations form tight deterministic bands; the states the policy actually visits are diffuse and lie largely between and around them. The two distributions are not disjoint, but the planner's density is concentrated where the policy is not.

Two limits apply. Coverage is measured on end-effector position, 3 of the 8 observed dimensions, so it says nothing about joint configuration. And it is descriptive rather than causal: the matched comparison establishes that the demonstration source mattered, while the coverage figure proposes how the two sources differ.

Observation aliasing, tested and rejected. The intuitive explanation for that failure is that the shield's corrections depend on state the policy cannot see: $\pi_{0.5}$ observes an image and eight proprioceptive dimensions but no joint angles, and a 7-DoF arm can place the end-effector identically with the elbow in very different positions — so two training samples could look identical while carrying contradictory targets, and behaviour cloning would average them. This was asserted early in the project and never measured, though it is measurable from the demonstration files alone: for each dataset, the disagreement between the target actions of observation-neighbours, normalised by the disagreement between random pairs, compared at matched observation *radius* rather than matched neighbour count, since a nearest-neighbour ratio conflates aliasing with sampling density and the planner's trajectories are far denser.

Figure 4.4 shows the prediction reversed. The planner's demonstrations are roughly twice as *well* determined by the observation at every radius: the teacher that fails to distil has the cleaner observation-to-action mapping, and the teacher that succeeds is the more aliased of the two. In hindsight the direction is unsurprising, since a scripted controller is close to a deterministic function of state — but the prediction was recorded before the planner figure was computed, and aliasing is not the binding constraint. Distribution shift and action-distribution mismatch remain as candidates.

Where the teacher had nothing to say. Six of the 24 scenes contain no planner demonstrations, so roughly 30 of the 120 evaluation rollouts test whether the expert could solve the scene at all rather than whether distillation worked. The gap is systematic rather

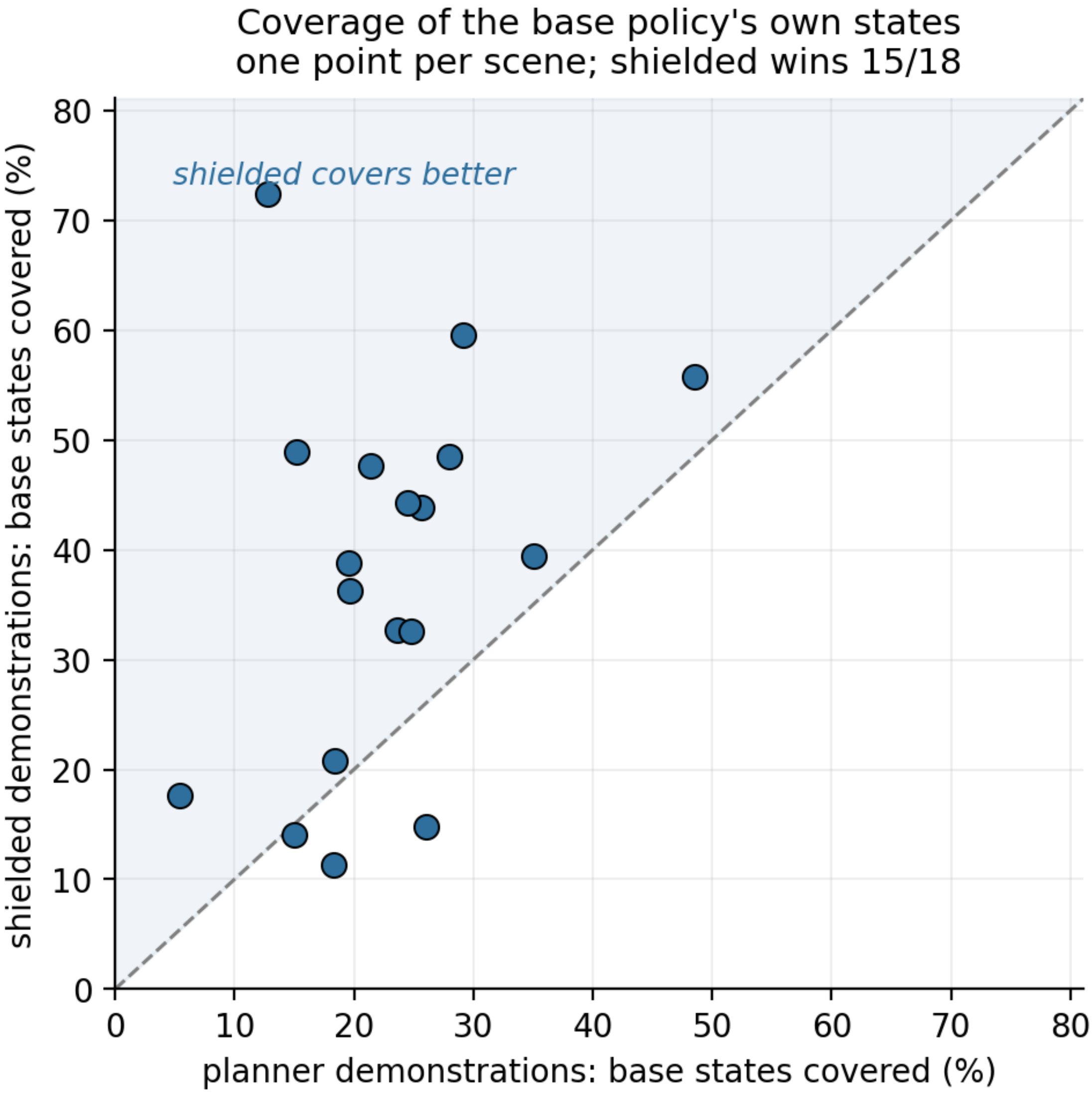

Figure 4.3: Both demonstration sets are subsampled to equal size per scene (30 resamples). **(a)** The shielded rollouts cover the policy's own states better in 15 of 18 scenes, 37.7% against 22.8% pooled. **(b)** The same comparison on the widest-gap scene. The planner's demonstrations form tight, near-deterministic trajectory bands, while the states the policy actually visits are diffuse and largely elsewhere.

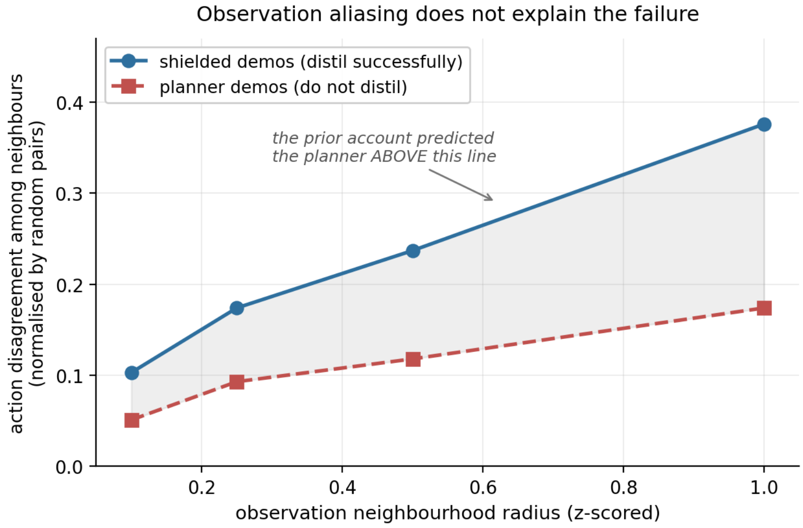

Figure 4.4: A prediction of this project, refuted by its own data. If the planner's corrections were poorly determined by what the policy observes, its curve would lie *above* the shielded one. It lies consistently below, by a factor of roughly two at every radius.

than random — the planner fails entirely on elevated-target tasks, a kinematic limit of the controller rather than a collision-avoidance one (Appendix B). Restricting to the 18 scenes with training data does not change the outcome, but both slices are worth distinguishing: the 24-scene figure conflates scenes where the teacher had nothing to say with scenes where it spoke and was not understood.

Answering RQ3, second part. The source of the demonstrations determines whether the transfer occurs. Corrections of equal density applied to a different controller's trajectories produced no measurable improvement, under a comparison in which the demonstration source was the only variable. The measured difference between the two sources is state coverage; observation aliasing was tested as an alternative explanation and rejected.

4.7 What Can the Shield Not Reach?

The shield reduces collisions to 13.3% but not to zero. This section identifies what the residual consists of, using per-body collision decomposition.

Table 4.6: Per-body attribution of collisions in 120 episodes.

Policy	Collided	Gripper	Arm link	Held object
Base, no shield	99	71	17	36
Base + shield	16	**0**	15	0
Round 1, no shield	23	7	13	9
Round 1 + shield	10	**0**	8	1
Round 2, no shield	21	8	8	7
Round 2 + shield	11	**0**	9	0

The split is the one Section 3.4 predicts. Where the barrier's model is precise and the constrained velocity is the commanded one — the end-effector — collisions are eliminated outright: not one gripper contact across 360 shielded episodes, so the discrete-time approximation of Section 3.5 holds in practice. Where the model is coarse and the velocity is inferred, they persist: arm links account for 15 of the 16 residual collisions in the shielded baseline.

This bears on how Section 4.4 should be read. The distilled policy's shield-free figure sits within a few points of what its teacher actually achieved, not of a perfect one — the teacher's ceiling is set by approximation error in the constraint rather than by an absence of authority.

The student exceeds its teacher. Arm-link collisions fall from 17 to 8 between the base and twice-distilled policies, while the shielded baseline — with those same constraints active — still records 15. The student is therefore not merely reproducing the filter's corrections: in the channel where the barrier is least able to enforce anything, it does better than the barrier does.

The likely route is the retention criterion rather than the correction signal. Demonstrations were kept only if nothing at all was displaced, arm links included, so the dataset encodes whole-body avoidance that no constraint row ever expressed. Safety reaches the policy through two channels — imitation of corrections where the filter has authority, and outcome selection where it does not — and only the second can carry behaviour the barrier cannot represent.

This is stated with less confidence than the gripper result: 17/120 against 8/120 is 14.2% [9.0, 21.5] versus 6.7% [3.4, 12.6], and those intervals overlap.

Answering RQ4, first part. The transfer does not cover the arm links. Where the barrier's model of the robot is precise and the constrained velocity is the one commanded, collisions are eliminated outright; where the model is coarse and the velocity inferred, they persist, accounting for 15 of the 16 residual collisions under the shield. The distilled policy nonetheless improves on its teacher in that channel, on overlapping intervals.

4.8 Are There Other Channels for Safety?

Four further mechanisms were tested. Table 4.7 summarises them.

Scalar-reward reinforcement learning

RL was tried first, and its failure motivated the imitation approach. The setup was flow-SDE GRPO on $\pi_{0.5}$'s flow-matching action head, with a LoRA adapter on the action head receiving gradients and the backbone frozen — the same trainable surface used for distillation in Section 4.4, so the comparison between the two learning signals is not confounded by capacity. Rollouts used ten denoising steps at noise 0.7 on one level-II scene, with

Table 4.7: Mechanisms tested and rejected, with the scope at which each negative is claimed. None is a general claim about its method class.

Mechanism	What was tested	Outcome	Scope of the negative
Scalar-reward RL	Flow-SDE GRPO on the action head, 4 configurations $\times$ 6 rounds, one scene	Either collapses to inaction or learns nothing; no setting does both	The reward design and budget tested. Not safe RL as a class; constrained-MDP and world-model methods supply richer signals
Language-expressed constraint	One safety clause appended to the instruction, $n = 60$ per condition, level II	No safety change; 23% slower, 11.7 points less successful	The single phrasing tested: generic, unnamed referent, manner adverbial
Best-of-N selection	$K \in \{4, 8\}$ candidates re-simulated per query, exact state rewind	No safe candidate on $\sim$60% of queries; doubling K buys 8.8 points	This policy and benchmark. The measurement it yields is reported as the result
Generative uncertainty as a runtime monitor	Three statistics of the stored flow-denoising chain, scored as AUC for predicting whether an episode collided	At chance for the base policy (0.461). The distilled policy inverts consistently (0.350, i.e. 0.65 flipped) but on 23 collision episodes	This policy and benchmark, and episode-level prediction. The inversion is roughly two standard errors from chance and is reported as suggestive

A further *explanation*, as opposed to a mechanism, was also tested and rejected: observation aliasing does not account for the transfer boundary (Section 4.5).

$K = 8$ rollouts per group over six rounds, and a reward combining success, direct collision, CBF activation rate and progress.

Two configurations bracket the failure. With the shield active on every rollout the only safety signal is the activation penalty, and the policy stayed flat even at a raised learning rate. Running half of each group's rollouts unshielded admits real collisions and widens the within-group spread to roughly $\Delta = 1.0$; safety improved immediately and the task collapsed with it, success reaching zero. These are not two settings with a working point between them: every configuration that moved the policy's behaviour collapsed it, and every configuration stable enough to preserve the task learned no avoidance. Section 5.5 takes up why.

Safety expressed in language

Both conditions share checkpoint, seed, held-out initial states and evaluation code, differing only in a safety clause appended to the instruction. Twelve level-II scenes at five initial states each give $n = 60$ per condition.

> **plain:** `pick up the black bowl between the plate and the ramekin and place it on the plate`
>
> **safety:** `... and place it on the plate, being careful not to touch or knock over any other objects`

The phrasing is generic: it names no object, gives no spatial reference, and states the constraint as a manner adverbial rather than a prohibition. Collisions were unchanged within interval overlap (70.0% against 68.3%), while success fell 11.7 points and episodes ran 23% longer. The policy responds to safety language with hesitancy rather than avoidance — a behavioural prior not connected to the geometry of the scene (Section 5.6).

Best-of-N selection

If safety varied across the actions a policy might sample from a given state, it could be obtained by rejection: draw K candidate action chunks, simulate each, execute the safest. The sampler must be stochastic for candidates to differ, so the matched control is $K = 1$ at the same noise level rather than the deterministic policy used elsewhere in this chapter. Making the measurement trustworthy required the simulator to be rewound exactly between candidates; snapshot and restore were verified to machine precision before any result was recorded.

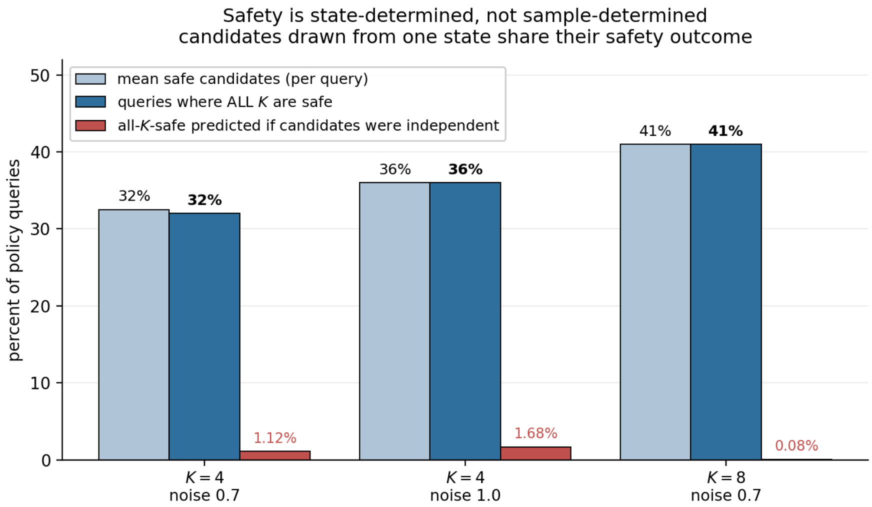

Figure 4.5: Why rejection sampling cannot help here. In every configuration the fraction of individually safe candidates coincides with the fraction of queries in which *all* K are safe, so a query is 0-of-K or all-of-K safe and almost never in between. Independent sampling would make the all-K-safe rate two to three orders of magnitude smaller (right-hand bars).

As a method this is negative: on roughly 60% of queries no candidate was safe, and doubling K from four to eight recovered 8.8 percentage points for twice the compute.

The measurement underneath it is the result. The mean safe fraction and the all-K-safe fraction coincide in all three configurations. Were the candidates independent draws the all-K-safe rate would be p^K: 1.1% at $K = 4$ and 0.08% at $K = 8$, against observed values of 32% and 41%. The policy's samples from a given state are therefore near-perfectly correlated in their safety outcome, and safety in this policy is **state-determined rather than sample-determined**. Once a state is reached the outcome is largely fixed, and no amount of choosing among actions recovers it — which is also why episode-level filtering worked (Section 4.5) where action-level selection cannot: episode-level selection chooses between trajectories that reached *different* states, the axis along which variation actually exists.

Generative uncertainty as a collision predictor

Uncertainty was also tested as a barrier-free monitor: $\pi_{0.5}$ stores the full denoising chain for every action, and if the policy were less certain where safe and unsafe behaviours are both plausible, that chain would carry a risk signal requiring no obstacle geometry at all. It does not — the association is at chance for the base policy, and the distilled policy's consistent *inversion* suggests, on limited evidence, that its remaining failures are confident ones.

Answering RQ4, second part. None of the four alternatives substituted for shielded imitation under the implementations and budgets tested. Each negative is scoped as stated in Table 4.7, and one of them yields a measurement rather than a method: within a given state, the safety outcomes of the policy's sampled action candidates are near-perfectly correlated.

4.9 Conclusion

This chapter reported the experiments in the order of the research questions. A control barrier function shield makes $\pi_{0.5}$ substantially safer on this benchmark, and reproduces the published baseline closely enough to validate the evaluation pipeline (RQ1). Distilling the shielded policy's own corrected rollouts internalises that safety: with no shield at inference the policy is both safer and more successful than the model it came from, recovering most of the shielded baseline's collision avoidance in a single round (RQ2).

Two controls bound that result. The improvement is attributable to the shield's corrections rather than to outcome selection, and it does not survive a change of demonstration source: a geometry-privileged planner supplying corrections of equal density produced no measurable gain (RQ3). The transfer leaves the arm links largely uncovered, and four alternative mechanisms — scalar-reward reinforcement learning, language-expressed constraints, best-of-N selection and uncertainty monitoring — failed to substitute for it (RQ4).

Chapter 5 interprets these findings, sets them against the literature of Chapter 2, and states what they do and do not establish.

Chapter 5

Discussion

5.1 Introduction

A vision-language-action policy that is competent at manipulation is not thereby safe, and the standard remedy leaves a control barrier function shield attached for the life of the deployment. This work asked whether those corrections can instead be compiled into the policy: whether the shield makes $\pi_{0.5}$ safe at all (RQ1), whether that safety survives the shield's removal (RQ2), what determines whether the transfer occurs (RQ3), and what it fails to cover (RQ4). The approach was a controlled comparison on SafeLIBERO — shield, collect, behaviour-clone, then evaluate with the shield detached — reported in Chapter 4 as measurements with their interpretation deferred. This chapter supplies that interpretation and sets it against the literature of Chapter 2. It follows the research questions in order: what the removal of the shield shows, what internalised caution costs, why the corrections transfer and from whom, and where the transfer stops.

5.2 Overview of Key Findings

The data support the claim that a barrier's corrections can be amortised into a policy's weights. Distilling $\pi_{0.5}$ on its own shielded rollouts left it substantially safer with no shield at inference, and more successful than the model it came from, on held-out initial states and with disjoint confidence intervals. A key pattern to emerge was that safety and competence moved together rather than trading off: every configuration that reduced collisions also raised or held task success, except when two safety mechanisms were stacked.

The analysis also identifies a boundary, and the boundary is the more transferable result. The improvement is attributable to the shield's corrections rather than to training on successful episodes, since an identically filtered set collected with the shield switched off left the policy no better than its baseline. Yet corrections of equal density supplied by a geometry-privileged motion planner transferred nothing at all. What separated the two teachers was not the density or the quality of their supervision but where in the state space it sat. Four alternative routes to the same target — scalar-reward reinforcement learning, language-expressed constraints, best-of-N selection and generative uncertainty monitoring

— each failed under the implementations and budgets tested.

5.3 What Removing the Shield Shows

The shield result itself is a replication and is treated as such: the unshielded baseline reproduces the figures AEGIS reports for the same policy on the same benchmark to within a point on both collision avoidance and task success [8], which is the evidence that the evaluation pipeline measures what it claims to. The finding that matters is what happened when the shield came off.

That result is easy to underrate, because the obvious comparison class cannot produce a counterexample to it. Methods that make a VLA safe by filtering its actions — AEGIS's barrier projection, ConBaT's learned barrier critic, LPB's latent predictive barrier [8, 10, 14] — leave the underlying policy's weights untouched. Removing the mechanism from any of them returns the baseline by construction, so the question of whether safety has been internalised is not one they neglect to ask; it is one their architecture cannot pose. Their silence is therefore no evidence either way.

The one published result that does shape a policy's weights while a barrier is active points the other way. Guerrier et al. train an agent under an active CBF filter, remove it, and find no internalisation — their critic assigns high value *inside* the obstacle [6]. Exposure to a shield during training is thus demonstrably insufficient on its own, which is what makes the present result a finding rather than a foregone conclusion, and which locates the cause in the training objective rather than in the shield's presence. Section 5.5 takes that up.

A second feature of the result needs interpreting, because it runs against the usual expectation that safety is bought with performance: task success *rose*, by a margin whose interval is disjoint from the baseline's. One reading is that collisions in this benchmark are not purely a safety cost. A displaced obstacle changes the scene the policy is conditioned on, and an episode that has knocked its target aside is frequently no longer completable, so avoiding collisions and completing tasks are partly the same objective. The obvious alternative reading — that the gain is ordinary self-improvement from imitating successful rollouts, with the shield incidental — is excluded by the matched control of Section 4.5, in which distillation on an identically filtered but unshielded set left success *lower* than the undistilled baseline. Filtering for success did not by itself teach success on this benchmark.

5.4 The Cost of Internalised Caution

The main cost is in execution time. The distilled policy is roughly 14% slower. It takes a more conservative route, and given the focus of this work in safety instead of performance, this is to be expected.

There is a cost trade-off that comes from combining mechanisms rather than of distillation itself. As shown in Figure 5.1, applying the shield to the distilled policy improves safety further but reduces success by similar amounts. When the distilled policy already

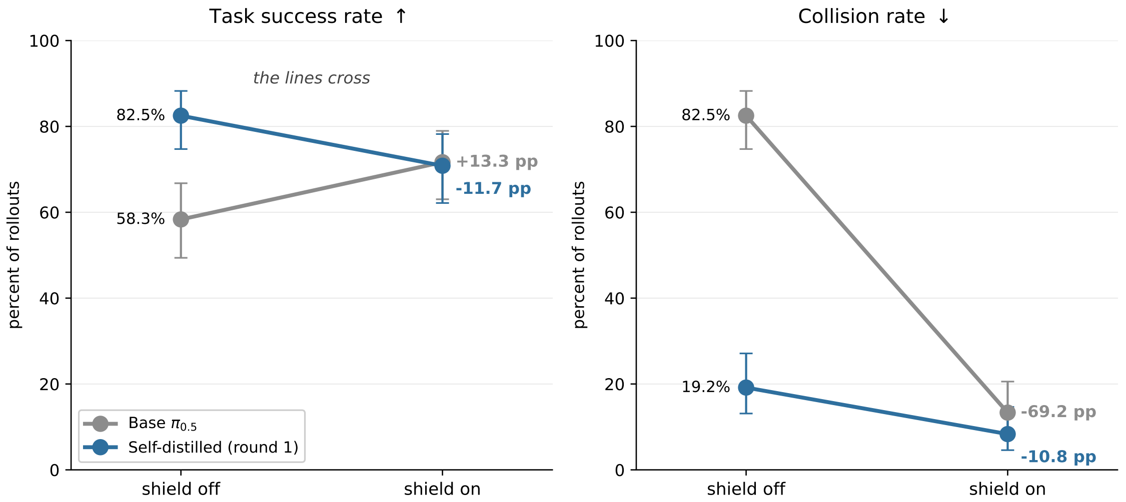

Figure 5.1: Attaching the shield raises the base policy's success and lowers the distilled policy's, while collisions fall in both cases: the reversal is confined to the success channel. $n = 120$ held-out rollouts per condition, Wilson 95% intervals.

avoids the obstacle, the shield's safe action projection perturbs actions that were already both safe and competent, leading to safety-induced distribution shift. This result is congruent to what is reported in VLSA, whose authors independently name the same effect and observe that enforcement can drive a policy out of distribution, where it "may behave erratically and fail to recover" [8]. That an approach built on runtime enforcement reports the ceiling of runtime enforcement is the strongest available argument for internalisation, and it reaffirms the relevance of self-distillation as an alternative route to safety that retains competence.

A second distillation round neither improved nor eroded performance, in contrast to an earlier DAgger-style experiment in this project that degraded across rounds (Section 4.6). One round was sufficient, and iterating was not harmful.

The trade runs both ways. Re-attaching the shield costs success but restores the online constraint satisfaction a formal safety argument requires, and it produces the highest collision-avoidance rate of any configuration tested — 91.7%, at a success rate statistically indistinguishable from shielding the base policy. Where a safety case demands a runtime constraint, distilling first is therefore the better starting point. That the residual is 8.3% rather than zero is a reminder that the guarantee is continuous-time and the implementation approximates it (Section 3.4).

5.5 Why the Corrections Transfer, and From Whom

Two routes to the same target were available — move the policy toward the behaviour the shield induces — and they differ only in the signal used. That difference turns out to be decisive, and for a structural reason rather than a tuning one.

A scalar reward communicates the target through one number per trajectory, leaving the policy to infer which of several hundred actions was responsible. In a flow-matching policy the only lever available for that inference is sampling noise, so exploration and credit assignment are the same knob: raising it enough to distinguish good actions from bad also degrades the actions being evaluated. Section 4.8 brackets the consequence from both sides, and no working point exists between them. Imitation removes the inference entirely. The shield does not report that a trajectory was unsafe; it reports what should have been done instead, at each step where it intervened, in the space the policy emits. Credit assignment is given rather than inferred, which is precisely what Guerrier et al.'s agent lacked: its actions were overridden while its objective remained a scalar reward, so no gradient ever pointed toward the corrected action [6].

What the Boundary Result Shows

Dense per-step supervision is necessary but not sufficient. The planner's targets were equally dense and equally corrected, and transferred nothing. What separated the two teachers was where their demonstrations sat: at matched sampling the planner covers the states the student reaches substantially less well (Figure 4.3).

The operative variable is therefore the state distribution the supervision covers, not the identity of the teacher — a distinction the literature supports independently. SAFE-GIL clones an MPC or PID expert and IntervenGen replays human corrections; both teachers are foreign to the learner, and both transfer safety, because both take care that the demonstrations sit at states the learner's errors would produce [3, 7]. Self-distillation is one route to that coverage rather than the only one. Its practical advantage is that it needs no disturbance model or human operator: the policy visits its own failure states unprompted.

This also sets the result against its closest conceptual neighbour. ROAD-VLA distils a corrective signal into a VLA and argues that the teacher must remain proximal to the current policy [16]; its teacher is a perturbed copy of the student itself, and every privileged alternative it rejects is a language template rather than a controller. Their proximity principle therefore *predicts* the planner's failure without ever testing a foreign low-level controller. The contribution here is the measurement they anticipate rather than a disagreement with them.

5.6 What the Transfer Does Not Reach

The transfer is uneven across the robot, and the pattern is informative rather than incidental. Collisions were eliminated where the barrier's model of the robot is precise and the constrained velocity is the one actually commanded, and persisted where the model is coarse and the velocity inferred — the arm links, which account for almost all residual collisions under the shield (Section 4.7). What distils is not avoidance in general but avoidance in the channel the barrier expresses well. This is consistent with the barrier being the source of the signal rather than a general safety prior, and it identifies the whole-body barrier as the specific engineering change most likely to raise the ceiling.

The four alternative mechanisms failed for reasons that are legible rather than mysterious, and two of them find independent support. The language channel produced no safety change and cost success, which matches ROAD-VLA's account of why their own text-guided variants underperformed: a modality gap prevents discrete text from supplying the precise grounding continuous control requires [16]. The reinforcement-learning negative reflects the poverty of a scalar signal in the sense set out above, and the approaches that do succeed on safety-constrained VLA training supply materially richer signals — explicit cost constraints in a constrained MDP [17], or an action-conditioned world model that predicts the consequence of a candidate before it is taken [15] — at a compute cost well beyond what was available here. Best-of-N selection yielded a measurement rather than a method: the near-perfect correlation among a state's sampled candidates says that in this policy, safety is a property of the state reached rather than of the action drawn, which is itself an argument for changing the policy rather than for filtering its samples.

5.7 Closing Summary

This chapter interpreted the findings in relation to the four research questions of Chapter 1. The results support the view that a control barrier function's corrections can be internalised into a vision-language-action policy, leaving it safer and more capable with the shield detached — a result that the filtering literature is structurally unable to contradict, and that the one comparable weight-shaping study argues against [6], which is what makes it worth reporting. Safety and competence moved together throughout, except where two mechanisms were stacked, an effect the primary baseline independently documents [8].

The study also identified a clear boundary. Dense, in-distribution corrections transfer; equally dense corrections from a geometry-privileged planner do not, and the variable that separates them is state coverage rather than the teacher's identity or competence. The transfer extends only as far as the barrier's model of the robot is faithful, and four alternative mechanisms did not substitute for it. These findings carry implications for how safety filters are deployed and for what remains unresolved, which the following chapter draws together.

Chapter 6

Conclusion

6.1 Introduction

This chapter begins with a summary of the key findings in relation to the research questions set out in Chapter 1, followed by the limitations that bound them and the recommendations for future research those limitations imply. It then states the contribution of the study, before closing with a brief overall recap.

6.2 Summary of Findings

This study aimed to determine whether the collision avoidance supplied by an inference-time control barrier function shield can be amortised into the weights of a vision-language-action policy, such that the shield is no longer required at deployment. Under the conditions tested, it can. In relation to RQ1, the shield made $\pi_{0.5}$ substantially safer on this benchmark and reproduced the published baseline closely enough to establish that the evaluation measures what it claims to. In relation to RQ2, distilling the shielded policy's own corrected rollouts reduced collisions from 82.5% to 19.2% **with no shield at inference**, while raising task success from 58.3% to 82.5% on held-out initial states — recovering most of the shielded baseline's collision avoidance in a single round, at a cost of roughly 14% in execution time.

In relation to RQ3, the transfer is bounded, and the bound is the more useful half of the result. The improvement is produced by the shield's corrections rather than by training on successful episodes, but it does not survive a change of demonstration source: a geometry-privileged motion planner, supplying corrections from the same shield, produced no measurable gain under a comparison in which the demonstration source was the only variable. What separates the teacher that worked from the one that did not is where its demonstrations sit. Transferable supervision must correct the learner at states the learner itself reaches.

In relation to RQ4, the transfer does not reach the whole robot. Collisions were eliminated where the barrier's model of the robot is precise and the constrained velocity is the one commanded, and persisted at the arm links, which account for almost all of what remains under the shield. Four alternative mechanisms — scalar-reward reinforcement learning, a language-expressed constraint, best-of-N selection and a generative uncertainty monitor — were each tested and none substituted for shielded imitation, though every one of those negatives is scoped to the implementation tested rather than to its method class.

6.3 Limitations

Simulation only. Every result is from SafeLIBERO. No physical robot was used and no claim of sim-to-real transfer is made.

No formal guarantee is inherited. The distilled policy carries no runtime constraint, so nothing of the shield's formal argument survives its removal. What is demonstrated is an empirical reduction in collisions.

Privileged geometry. The barrier is constructed from obstacle poses and meshes read directly from the simulator, so every shielded figure reported here is an upper bound for this class of filter. The distilled policy uses no geometry at inference, but the dependency remains during training.

Approximate robot model. The barrier represents the robot with spheres rather than its exact shape, because the exact version would not solve fast enough to run at every control step (Section 3.4). How accurate that approximation actually was is never measured directly.

Single training seed. Each distilled policy is one training run and seed variance is unquantified.

Single policy and action space. Everything here uses one model, $\pi_{0.5}$, which produces chunks of actions by integrating a flow field.

Scope of the negative results. Each of the four negatives is scoped to the implementation and budget tested rather than to its method class, as stated in Table 4.7.

Attribution caveats. The per-body attribution is a documented lower bound that can miss delayed or indirect contacts.

6.4 Recommendations for Future Research

A physical robot is the most direct extension. Whether internalised avoidance survives contact dynamics, calibration error and a real perception pipeline is the question this work most needs answered, and Section 5.4 suggests the deployment configuration to test: the distilled policy with the shield reattached, which reached the highest collision-avoidance rate of any arm evaluated while keeping the runtime constraint a safety case would require.

Repeating the training across seeds is the cheapest outstanding item and addresses the weakest one. Each distilled policy here is a single run, so the training variance behind the headline is unquantified even though the evaluation variance is not.

Perception-grounded barriers would remove the privileged-geometry dependency and measure how much of the reported safety survives estimation error — the same question a physical deployment asks, isolated from the rest of the reality gap. Learned safety representations offer one route [10, 14], though both retain the inference-time correction this work removes.

A whole-body barrier with proper link geometry and directly constrained link velocities would lower the shield's own floor (Section 4.7); whether that improvement then distils is separately testable. It would also make CBF-guided sampling worth testing, since the existing implementation was verified against a closed-form projection but wired to an end-effector barrier, and so never activated on the scenes where arm links collided.

Other architectures and moving obstacles bound the generality of the result. The best-of-N and reinforcement-learning findings depend on chunked flow sampling and may not hold for policies emitting single actions, and every scene here is static — the case that matters for deployment is a constraint that moves.

Finally, **separating the surviving explanations** would say something general about when demonstrations transfer between controllers. Observation aliasing was measured and rejected (Section 4.6) and state coverage is supported but measured only on end-effector position; distribution shift and action-distribution mismatch remain. A direct comparison against model-based safe RL on this benchmark [15] would place distillation against the strongest current alternative.

6.5 Contribution of the Study

Two neighbouring claims would be false and are not made. Learning safety from CBF-related supervision is not new; it has been treated theoretically [4], and learning from CBF demonstrations in simulation predates that work. Distilling a corrective signal into a vision-language-action policy is not new either [16].

What is new is the conjunction: a control-theoretic safety filter, distilled from a VLA's own shielded rollouts and evaluated with the filter *removed*, in the setting where runtime shielding is the current state of the art. With it comes the boundary condition. That condition is narrower than the claim it is easily mistaken for, because foreign teachers can transfer safety: SAFE-GIL clones an MPC or PID controller and IntervenGen replays human corrections [3, 7], and both succeed because both take care that their demonstrations sit where the learner's errors would put it. The planner tested here received no such treatment. Policy-proximal self-distillation predicts precisely this outcome [16], but every teacher that work rejects is textual and every teacher it accepts is derived from the student, so no foreign low-level controller appears anywhere in it. Supplying one under a matched comparison, and measuring what follows, is this thesis's contribution.

Three findings bear directly on how such a system would be deployed. Distilling before shielding produces the lowest collision rate of any configuration measured, at a success rate indistinguishable from shielding the base policy, so where a runtime constraint is required it is better applied to a distilled policy than to a raw one. The distilled policy also invokes the shield roughly half as often — the barrier alters 18% of control steps against 36%

before distillation — which matters wherever motion must be predictable, since a filter that fires constantly is a filter that makes cycle times vary. And safe demonstrations are approximately twice as cheap to collect with the shield active as without it, at a 39% yield against 18%, a direct saving for any deployment gathering task-specific data.

A further contribution is a measurement rather than a method: within a given state, the safety outcomes of a VLA's sampled action candidates are near-perfectly correlated, which is why action-level rejection sampling cannot help where episode-level selection can (Section 4.8).

6.6 Closing

Vision-language-action policies are competent manipulators and unsafe ones, and the prevailing remedy keeps a control barrier function attached for the life of the deployment. This study asked whether that filter's corrections could instead be compiled into the policy's weights, and found that they can: distilled on its own shielded rollouts, $\pi_{0.5}$ became both safer and more capable with nothing running at inference, and the corrections transferred only when they had been applied to states the policy itself reaches.

The practical conclusion is narrower than "safety can be learned". A runtime filter teaches a policy the regularities it observes and corrects, and those corrections amortise into weights without the per-step optimisation comparable learned-barrier methods retain at inference [10, 14], and at a fraction of their training cost. But the learned policy is only as broad as the states, hazards and geometry those corrections represent. Runtime shielding remains necessary outside that envelope. Distillation reduces dependence on the shield; it does not remove the need for a safety case.

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Appendix A

Additional results

Interactive distillation, and the diagnostic that located its failure. Before the offline comparison reported above, the classical expert was distilled interactively in the DAgger manner: the student drives, the expert labels the states the student visits, and the aggregated dataset is retrained. This is the textbook remedy for exactly the coverage problem identified below, and it made matters worse. Collision rose monotonically across rounds, from 44% to 75%, with success pinned near zero.

The cause was not the coverage principle but a violated precondition. DAgger's correctness argument requires a Markov oracle $\pi^*(s)$: the expert's label must be a function of the state it is shown. The classical expert was implemented as a latched phase machine carrying hidden internal state, so when the student drove erratically the expert's phase desynchronised from the observation and the same image received contradictory labels across rounds. Behaviour cloning averaged them.

Isolating this required ruling out the alternative, that the imitation machinery itself was broken. Hard-overfitting behaviour cloning on a single clean demonstration and evaluating on that same episode returned success 1.0 with zero collisions, establishing that normalisation, the observation pipeline and adapter capacity were all sound, and that the fault lay in the labels. The expert was then rewritten as a stateless reactive oracle, inferring its phase from observables at every call — grasp from the measured gripper finger separation, transport from end-effector height — and validated against the original on the same episodes, where it produced identical outcomes episode-for-episode.

The rewrite made the expert Markov, and distillation still failed. That is what moved the investigation from the interactive protocol to the teacher itself, and ultimately to the matched offline comparison reported above. It is recorded here because the negative is easy to misattribute: an interactive method failing against a non-Markov expert is a statement about the expert, not about DAgger or about coverage.

Appendix B

The Classical Controller Characterised on Its Own

Section 4.6 reports that six of the 24 scenes contain no planner demonstrations and that the gap is systematic. This appendix gives the evidence, from the full-grid collection of 24 scenes $\times$ 36 attempts = 864 rollouts under ground-truth geometry. A demonstration is *clean* if it succeeded and displaced no obstacle; the overall yield is 318/864, or 37%.

B.1 The Yield Is Bimodal, Not Uniformly Mediocre

Table B.1: Clean demonstrations per scene, out of 36 attempts. Two scenes exceed 94%; six produce nothing at all, so the aggregate of 37% misrepresents both ends of the distribution.

Scene	Clean	Scene	Clean
object_LI_t3	35 (97%)	spatial_LI_t2	17 (47%)
object_LII_t1	34 (94%)	spatial_LI_t1	14 (39%)
goal_LI_t0	26 (72%)	object_LII_t0	14 (39%)
goal_LI_t1	26 (72%)	spatial_LII_t2	14 (39%)
goal_LI_t2	22 (61%)	spatial_LI_t0	12 (33%)
spatial_LII_t3	20 (56%)	object_LII_t2	12 (33%)
object_LII_t3	19 (53%)	spatial_LI_t3	11 (31%)
goal_LII_t2	17 (47%)	goal_LII_t0	9 (25%)
		object_LI_t0	8 (22%)
		goal_LII_t3	8 (22%)
Zero yield: goal_LII_t1, goal_LI_t3, object_LI_t1, object_LI_t2, spatial_LII_t0, spatial_LII_t1			

The failure is not a difficulty effect. Three of the six zero-yield scenes are level I, while spatial_LII_t3 reaches 56% and object_LII_t1 reaches 94%. Competence is task-specific rather than level-specific, which rules out “level II is simply harder” as the explanation.

B.2 Three Separable Failure Mechanisms

Phase occupancy from the per-step logs of the six zero-yield scenes separates three causes rather than one. Counts measure time spent in a phase rather than terminal outcome.

Table B.2: Dominant execution phases in the six zero-yield scenes, with the two highest-yield scenes for contrast. Reaching PLACE is what distinguishes success.

Scene	Dominant phases	Mechanism
goal_LII_t1	PLAN_FAILED 3150 (no other phase reached)	planning infeasibility
goal_LI_t3	DONE 1553, TRANSPORT 394	task inexpressibility
object_LI_t1	APPROACH 1509, TRANSPORT 714 (no PLACE)	execution
object_LI_t2	APPROACH 1228, TRANSPORT 1022, PLACE 22	execution
spatial_LII_t0	APPROACH 1437, DONE 545	execution
spatial_LII_t1	DONE 1214, APPROACH 495	execution
object_LI_t3	APPROACH 413, TRANSPORT 296, PLACE 65	—
object_LII_t1	TRANSPORT 673, APPROACH 420, PLACE 78	—

Planning infeasibility. In goal_LII_t1, PLAN_FAILED is the only phase ever reached: RRT together with the inverse-kinematics and clearance ladder never returns a collision-free plan, and execution is never attempted. This is purely geometric.

Task inexpressibility. In goal_LI_t3 the plan executes to completion and the task is still not satisfied. The task is *open the top drawer and put the bowl inside* — two-stage manipulation that a pick-and-place waypoint sequence cannot represent. Zero yield here is a statement about what was implemented, not about the method.

Execution and tracking failure. In the remaining four scenes a plan exists but the arm does not track it to placement: time concentrates in APPROACH and TRANSPORT while PLACE is barely or never reached. The two highest-yield scenes are distinguished precisely by reaching PLACE.

B.3 Seven Iterations, and Why the Residual Is Structural

Table B.3: Controller iterations, each with a measured diagnosis rather than a guess. The aggregate never leaves the 35–40% band.

Version	Change	Clean ($n = 48$)
v2	clearance inflation	40%
v3	clearance ladder + IK retry	35%
v4	fallbacks	29%
v5	payload speed limit (object slipping from grip)	38%
v6	scene settling before measurement	38%
v8	four fixes from visual inspection	35%
v10	orientation gate on waypoint arrival	**40%**

Individual fixes are real and diagnosed — the orientation gate took a closing error from 35.5° to 0.0° on a scene that had regressed to zero — but each trades scenes against one another rather than lifting the total. That pattern, across seven attempts, is the evidence that the residual is structural rather than a tuning deficit.

B.4 A Physical Constraint Mistaken for a Control Failure

One earlier failure is worth recording for how it was diagnosed. Several bowl tasks sat at 0% success with the end-effector stalling roughly 5 cm above the object centre, which reads as a controller or inverse-kinematics problem. It was neither: the akita bowl is 11 cm wide and the gripper opens to 8 cm, so a top-down straddle grasp is physically impossible and no controller could have solved it.

The fix was a change of grasp strategy rather than of control. The gripper's fingers close along the world y axis, so offsetting the grip point by 5 cm along that axis drops one finger inside the bowl and one outside, pinching the rim on close. Bowls are selected for this mode by name while cartons retain the top-down grasp, and the full three-dimensional grasp offset is tracked so that placement drives the *object* to the goal rather than the offset end-effector. A speed-capped, xy-locked descent removes operational-space coupling drift. The spatial suite went from 0% to 69% with zero collisions.

This bears on how Section 4.6 should be read. The teacher's remaining failures were investigated for this class of explanation — a physical constraint masquerading as a control deficit — and the elevated-target failures survived that check, which is what makes the kinematic-limit reading credible rather than merely convenient.

B.5 What This Does and Does Not Explain

With full privileged geometry the controller reaches 37% clean demonstrations, bimodally distributed, failing through three separable mechanisms of which one is outside the scope of what was built and one is purely geometric. A task-and-motion planner with drawer primitives would address task inexpressibility; it would not obviously address execution failure, where most of the lost yield sits.

Neither would change the distillation result. Section 4.6 shows the planner student is statistically indistinguishable from the undistilled base policy even when restricted to the 18 scenes where demonstrations exist, so the outcome does not turn on the teacher's yield. What separates the two teachers is the state distribution their demonstrations occupy, not the number of scenes either could solve.

Appendix C

Implementation notes